A Stackelberg Game for Multi-Objective Demand Response in Dynamic Economic Emission Dispatch

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Abstract

This paper integrates game-theoretic demand response with multi-objective Dynamic Economic Emission Dispatch (DEED) in a smart grid framework. The proposed Smart Grid Stackelberg DEED (SGSD-DEED) model formulates a multi-period Stackelberg game in which the utility company optimizes generator dispatch and incentive payments while customers respond with curtailment, demand shifting, energy storage, local generation, and bidirectional grid exchange. Equilibrium existence is established via backward induction under standard convexity assumptions. The tri-objective formulation minimizes fuel cost and emissions while maximizing customer utility, solved using interior-point methods. Experiments with 5–50 customers and 6–10 generators demonstrate reductions of up to 56% in cost, 77% in emissions, and 83% in losses compared to baseline DEED. Validation on the IEEE 30-bus system with DC power flow constraints confirms that transmission limits add only 3–5% cost overhead. A Saudi Eastern Province case study with time-of-use pricing and gas-dominated generators achieves 46–47% cost and 68–70% emission reductions, confirming regional applicability. Sensitivity analysis across 64 weight combinations shows cost robustness (coefficient of variation CV = 1.5%), and Pareto front analysis via ε-constraint confirms modest cost–emission trade-offs (4.5% cost increase for 28% emission reduction). The model scales to 50 customers with strong per-customer economies.

Keywords: Multi-objective optimization, dynamic economic emission dispatch, Stackelberg game, demand response, energy storage, smart grid, DC optimal power flow.

1 Introduction

Dynamic Economic Emission Dispatch (DEED) is an optimization problem in power systems that seeks to balance electricity supply and demand while minimizing both fuel costs and pollutant emissions. DEED extends the Dynamic Economic Dispatch (DED) problem by integrating emission objectives alongside cost minimization [1]. Demand Response (DR) complements DEED by managing electricity consumption patterns on the customer side, and is widely recognized as an energy-saving measure [2] that can lower electricity bills [3].

The integration of DR with DEED enables the simultaneous optimization of demand-side load reduction and supply-side generator dispatch. DR-DEED programs have been proposed to manage both the operation of thermal generators and the electricity consumption patterns of consumers [1].

Effective implementation of DEED with DR requires smart grid (SG) infrastructure that supports real-time monitoring, bidirectional communication, and intelligent control. Smart grids utilize advanced digital and

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communication technologies to manage electricity flow from generation to consumption, and are particularly important for integrating variable renewable energy sources. Demand Response Management (DRM), a key component of smart grids, shifts non-essential consumption from peak to off-peak periods, reducing peak demand and lowering consumer costs.

1.1 Literature Review

We organize the review into three thematic areas: game-theoretic demand response, multi-objective DEED methods, and network-constrained dispatch.

Early game-theoretic formulations for smart grid demand management include the Nash equilibrium pricing models of Cui *et al.* [4] and the non-cooperative Demand Side Management (DSM) framework with distributed energy resources of Atzeni *et al.* [5, 6]. Belhaiza and Baroudi [7] introduced a multi-agent game for smart grid demand regulation, and Belhaiza *et al.* [8] extended this to a multi-periodic game with demand shifting and multi-objective optimization. More recent works have advanced the Stackelberg framework for DR. Cui *et al.* [9] combined Long Short-Term Memory (LSTM)-based load forecasting with a Stackelberg pricing game. Fochesato *et al.* [10] designed a bilevel incentive-based DR scheme for prosumers with renewables and storage, solving it via Karush-Kuhn-Tucker (KKT) reformulation with convergence guarantees. Li *et al.* [11] proposed a Stackelberg-evolutionary joint game for multi-microgrid DR and proved equilibrium existence and uniqueness. Wan *et al.* [12] constructed a Stackelberg–Nash game for price-based DR in retail markets, establishing equilibrium uniqueness via a distributed iterative algorithm. Wu *et al.* [13] formulated a Stackelberg–Nash game for PV prosumer coordination in microgrids under price-based DR. Xue *et al.* [14] proposed a bilevel Stackelberg model for multi-integrated energy systems with shared storage, using a non-dominated sorting whale optimization at the upper level. Ali *et al.* [15] developed a Stackelberg framework for priority-aware dynamic pricing and load scheduling in smart grids. These recent works demonstrate the maturation of formal Stackelberg frameworks for DR, but none integrates the game-theoretic structure with the DEED problem.

The seminal work of Nwulu and Xia [1] integrated game-theoretic DR into DEED, using a single-period Stackelberg model to determine optimal incentives for customer load curtailment. Ullah *et al.* [16] extended DEED to a tri-objective DSM formulation with wind energy penetration. Nalini *et al.* [17] applied an improved multi-objective moth flame optimizer (MFO) with a crowding distance mechanism for DEED with transmission loss prediction. Soni and Bhattacharjee [18] incorporated renewable energy and plug-in electric vehicles into DEED using an equilibrium optimizer. Recent advances include Shahri *et al.* [19], who addressed multi-area DEED with battery storage and renewables using a hybrid Particle Swarm Optimization (PSO)-Whale optimization on a 40-unit four-area network. Guo [20] formulated an economic dispatch model for renewable energy systems with demand response, incorporating wind and solar uncertainty via differential evolution. These DEED studies predominantly employ metaheuristic solvers and weighted-sum scalarization, but lack formal game-theoretic structure, equilibrium analysis, and expanded customer flexibility mechanisms (e.g., demand shifting, local generation, bidirectional grid exchange).

Several recent studies have explored multi-level market structures and network-aware dispatch. Kafshian and Saniee Monfared [21] proposed a tri-level market model with distribution companies, microgrids, and DR aggregators as distinct players. Ma and Venkatesh [22] co-optimized a DR market with the conventional energy market via optimal power flow (OPF), demonstrating significant cost reductions on Pennsylvania-New Jersey-Maryland (PJM) system data. Chen [23] developed a bilevel carbon-aware DR dispatch model validated on the IEEE 39-bus system, incorporating nodal carbon emission flow into the OPF formulation. Wang *et al.* [24] provided a comprehensive review comparing economic dispatch, DC Optimal Power Flow (DC-OPF), and AC-OPF formulations on IEEE benchmarks, highlighting the importance of network constraints for realistic dispatch.

While the above literature addresses game-theoretic DR and multi-objective DEED separately, a unified framework that combines (i) a formally defined multi-period Stackelberg game with equilibrium analysis, (ii) expanded customer flexibility including demand shifting, storage, local generation, and bidirectional grid exchange, (iii) network-aware dispatch via DC power flow constraints, and (iv) validation across diverse regional grid contexts has not been proposed. This paper addresses this gap.

1.2 Key Contributions and Novelties

While prior work has addressed game-theoretic demand response [8] and game-theoretic DEED [1] separately, their integration into a unified multi-period framework with expanded customer flexibility has not been formalized. We introduce the SGSD-DEED model, which combines multi-objective DEED optimization with a formally defined Stackelberg game for smart grid demand response. Table 1 summarizes the key distinctions between this work and its two primary antecedents.

Table 1: Comparison of the proposed SGSD-DEED model with prior game-theoretic DEED models.

Feature	**[1]**	**[8]**	**This paper**
Game type	Stackelberg (single-period)	Multi-period game	Multi-period Stackelberg
DR mechanism	Curtailment only	Curtailment + shifting	Curtailment + shifting + storage + DG*
Objective	Bi-objective (cost, emission)	Customer utility	Tri-objective (cost, emission, utility)
Customer actions	Reduce load	Reduce, shift	Reduce, shift, store, produce, sell
Equilibrium analysis	None	None	Formal SPNE with existence proof
Grid model	Power balance	Smart grid	Smart grid with flow conservation
Test scale	≤ 5 customers	≤ 5 customers	5–50 customers
Regional validation	None	None	PJM + Saudi Eastern Province

*DG = Distributed Generation.

(i) We formalize the SGSD-DEED as a multi-period Stackelberg game (Definition 1) with explicit players, strategy spaces, and payoff functions. We define the Subgame Perfect Nash Equilibrium (Definition 2) and prove its existence under standard convexity assumptions (Proposition 1). Prior models [1, 8] lack formal game specification and equilibrium analysis.

(ii) Unlike [1] (curtailment only) and [8] (curtailment and shifting), our model enables customers to simultaneously curtail, shift, store, produce, and sell energy, captured through nine classes of decision variables within a tri-objective optimization framework. The formulation incorporates DC power flow constraints to enforce transmission capacity limits and per-bus nodal balance.

(iii) We demonstrate reductions of up to 53% in fuel cost, 72% in emissions, and 80% in system losses compared to the DR-DEED model [1], and validate scalability with up to 50 customers and 10 generators.

(iv) We validate the model on a Saudi Eastern Province case study with a representative demand profile (air-conditioning–driven peak), gas-dominated generators, and a constructed time-of-use tariff (4:1 peak ratio), confirming that the framework generalizes across grid contexts without algorithmic recalibration.

2 Basic Models for DEED of Electric Power Generation

DEED and DR-DEED have been solved using mathematical optimization and metaheuristic methods. Early studies integrating DEED with DR used soft computing algorithms [25–28]. Arif and Javed [29] used a genetic algorithm to integrate economic dispatch and demand-side management in a micro-grid context. Nwulu and Xia [1] combined DEED with a game theory-based DR program accommodating multiple customers and time intervals. Chinnadurrai and Victore [30] addressed the DEED problem as a multi-objective optimization within a smart grid framework, incorporating wind power variability and a DR program that adjusts consumption across demand periods.

2.1 Basic DEED Formulation

The DEED problem integrates both DED and emission dispatch into a single multi-objective optimization problem [31]. The objective is to find trade-off solutions that minimize fuel cost and emissions simultaneously while satisfying network constraints. Table 2 details the mathematical notations used.

Considering C_j^t and E_j^t, respectively, as the fuel and the emissions costs for generator j at time t, the problem can be formulated mathematically as follows:

$$\min \sum_{t\in T} \sum_{j\in J} C_j^t \left(q_j^t\right) \tag{1}$$

$$\min \sum_{t\in T} \sum_{j\in J} E_j^t \left(q_j^t\right) \tag{2}$$

Symbol	Type	Description
q_j^t	Variable	Power generated from generator j at time t.
a_j, b_j, c_j	Parameter	The fuel cost coefficients of generator j respectively.
e_j, f_j, g_j	Parameter	The emission cost coefficients of generator j respectively.
J and T	Set	The set of generators and the dispatch time interval respectively.
D_t	Real	Total system demand at time t.
loss_t	Real	Total system losses at time t.
$q_j^{\min}$ $q_j^{\max}$	Real	Minimum and maximum capacity of generator j respectively.
DR_j UR_j	Real	Maximum ramp down and up rates of generator j respectively.
$B_{j,k}$	Matrix	The (j,k)-th element of the loss coefficient matrix of dimension $\lvert J\rvert \times \lvert J\rvert$.

Table 2: Summary of Basic Mathematical Notations

with

$$C_j^t\left(q_j^t\right) = a_j + b_j q_j^t + c_j (q_j^t)^2, \quad \forall j \in J,\ \forall t \in T, \tag{3}$$

$$E_j^t\left(q_j^t\right) = e_j + f_j q_j^t + g_j (q_j^t)^2, \quad \forall j \in J,\ \forall t \in T. \tag{4}$$

subject to the following network constraints:

$$\sum_{j\in J} q_j^t = D_t + \text{loss}_t,\ \forall t \in T, \tag{5}$$

$$q_j^{\min} \le q_j^t \le q_j^{\max},\ \forall j \in J,\ \forall t \in T, \tag{6}$$

$$-DR_j \le q_j^{t+1} - q_j^t \le UR_j,\ \forall j \in J,\ \forall t \in T, \tag{7}$$

where

$$\text{loss}_t = \sum_{j\in J}\sum_{k\in J} q_j^t B_{j,k} q_k^t,\ \forall t \in T. \tag{8}$$

The multi-objective problem is converted to a single objective via weighted scalarization with weight w:

$$\min \left[w \left[\sum_{t\in T}\sum_{j\in J} C_j\left(q_j^t\right)\right] + (1-w)\left[\sum_{t\in T}\sum_{j\in J} E_j\left(q_j^t\right)\right]\right] \tag{9}$$

2.2 Basic Game-Theoretic DR-DEED Formulation

Game theory provides a natural framework for modeling demand response, where incentive payments to customers must outweigh the costs they incur from service interruptions. Fahrioglu and Alvarado [32] introduced game-theoretic demand contract formulations for a single customer. Nwulu and Xia [1] extended this to multiple customers and time intervals, integrating the game-theoretic DR model into the DEED problem.

By considering the new mathematical variables and parameters in Table 3, we can define the Demand Response-DEED (DR-DEED) model.

For more than one customer i and more than one period of time t, the cost incurred by a customer of type θ_i who decreases power consumption by χ_i^t MW, represented by $c(\theta_i, \chi_i^t)$, can be defined as:

$$c(\theta_i, \chi_i^t) = K_{1i} \times (\chi_i^t)^2 + K_{2i} \times \chi_i^t(1-\theta_i),$$

In this cost function:

- $K_{1i} \times (\chi_i^t)^2$: represents a quadratic discomfort cost that increases with the amount of consumption reduction.
- $K_{2i} \times \chi_i^t(1-\theta_i)$: captures the *discomfort penalty* modulated by customer willingness $\theta_i \in [0,1]$. A highly willing customer ($\theta_i \to 1$) faces a vanishing penalty since $(1-\theta_i) \to 0$, making curtailment inexpensive. Conversely, an unwilling customer ($\theta_i \to 0$) faces the full penalty $(1-\theta_i) \to 1$, reflecting high discomfort from load reduction.

Symbol	Type	Description
θ_i	Real	Customer type variable, normalized in the range $[0, 1]$. It represents the willingness of customers to participate, where 0 indicates no interest in participating and 1 indicates a high willingness to participate.
χ_i^t	Real	Power reduction by participating customer i at time t.
ω_i^t	Real	Monetary compensation received by customer i at time t.
λ_i^t	Real	Power interruptibility value, representing the cost of not supplying power to a specific location on the grid.
K_{1i}, K_{2i}	Real	The cost coefficients.

Table 3: Game-Theoretic DR-DEED Mathematical Notations

The marginal cost is $\frac{\partial c}{\partial \chi_i^t} = 2K_{1i}\chi_i^t + K_{2i}(1 - \theta_i)$, which is non-negative and increasing in χ_i^t, confirming the convexity of the cost function. Customers with higher willingness ($\theta_i \to 1$) have lower marginal cost and thus higher marginal benefit from participation, while customers with lower willingness ($\theta_i \to 0$) have higher marginal cost and lower marginal benefit. Note that $c(\theta_i, 0) = 0$, ensuring zero cost when no curtailment occurs. The customer utility function (benefit) is derived from:

$$V_1(\theta_i, \chi_i^t, \omega_i^t) = \omega_i^t - c(\theta_i, \chi_i^t), \quad i = 1, \ldots, I.$$

Notice that the customer would participate if $V_1 \geq 0$. The utility benefit is determined by:

$$u_0 = \sum_{i \in I} \lambda_i^t \chi_i^t - \omega_i^t.$$

The objective is to maximize the expected utility benefit:

$$\max_{\chi, \omega} \sum_{t \in T} \sum_{i \in I} \left(\lambda_i^t \chi_i^t - \omega_i^t\right)$$

subject to:

$$\begin{aligned}
&\sum_{t \in T} \left[\omega_{i,t} - \left(K_{1,i}\chi_{i,t}^2 + K_{2,i}\chi_{i,t} - K_{2,i}\chi_{i,t}\theta_i\right)\right] \geqslant 0, \text{ for } i = 1, \ldots, I. \\
&\sum_{t \in T} \left[\omega_{i,t} - \left(K_{1,i}\chi_{i,t}^2 + K_{2,i}\chi_{i,t} - K_{2,i}\chi_{i,t}\theta_i\right)\right] \\
&\quad \geqslant \sum_{t \in T} \left[\omega_{i-1,t} - \left(K_{1,i-1}\chi_{i-1,t}^2 + K_{2,i-1}\chi_{i-1,t} - K_{2,i-1}\chi_{i-1,t}\theta_{i-1}\right)\right], \\
&\quad \text{for } i = 2, \ldots, I. \\
&\sum_{t \in T} \sum_{i \in I} \omega_{i,t} \leqslant UB, \\
&\sum_{t \in T} \chi_{i,t} \leqslant CM_i, \qquad \text{for } i = 1, \ldots, I.
\end{aligned}$$

where UB is the utility's total budget and CM_i is the daily limit of interruptible power for customer i.

2.3 Integrated DEED and Multi-Periodic Game-Theoretic Formulation

Nwulu and Xia [1] combined the DEED problem with the game-theoretic DR model into a single-objective DR-DEED formulation:

$$\min \left[w_1 \left[\sum_{t \in T} \sum_{j \in J} C_j\left(q_j^t\right) \right] + w_2 \left[\sum_{t \in T} \sum_{j \in J} E_j\left(q_j^t\right) \right] - w_3 \left[\sum_{t \in T} \sum_{i \in I} \left[\lambda_i^t \chi_i^t - \omega_i^t\right] \right] \right]$$

where w_1, w_2 and w_3 are the weights satisfying $w_1 + w_2 + w_3 = 1$. This formulation determines the optimal generator output while the game-theoretic DR component identifies the optimal customer load to curtail and the corresponding incentive payments.

3 The Proposed SGSD-DEED Model

The SGSD-DEED model extends the DR-DEED framework by integrating demand shifting, energy storage, and distributed generation into a multi-period game-theoretic optimization.

3.1 Game-Theoretic Framework

The SGSD-DEED model is formulated as a multi-period Stackelberg game, in which the utility company acts as the leader and customers act as followers.

Definition 1 (Multi-Period Stackelberg Game)**.** *The SGSD-DEED problem is modeled as a dynamic Stackelberg game*

$$G_{SGSD} = \{\mathcal{N},\ \mathcal{S}_{Leader},\ \{\mathcal{S}_i^{Follower}\}_{i\in I},\ \mathcal{U}_{Leader},\ \{\mathcal{U}_i\}_{i\in I}\} \tag{10}$$

with components:

- Players ($\mathcal{N}$)*: The leader is the utility company (grid operator), and the followers are n customers indexed by $i \in I = \{1, 2, \ldots, n\}$.*
- Leader strategy space*: The utility company selects generator dispatch levels and incentive payments:*

$$\mathcal{S}_{Leader} = \big\{(q_j^t, \omega_i^t)_{j\in J,\, i\in I,\, t\in T} : q_j^{\min} \le q_j^t \le q_j^{\max},\ \omega_i^t \ge 0,\ \textstyle\sum_{i\in I}\sum_{t\in T}\omega_i^t \le UB\big\}. \tag{11}$$

- Follower strategy space*: Each customer i selects curtailment, production, storage, shifting, and grid exchange decisions:*

$$\mathcal{S}_i^{Follower} = \big\{(\chi_i^t, p_i^t, x_i^t, s_i^t, h_i^t, f_i^t)_{t\in T} \in \mathbb{R}_+^{6T} : \textit{Constraints } (17)\text{–}(30) \textit{ hold for customer } i\big\}. \tag{12}$$

- Leader payoff *(to minimize):*

$$\mathcal{U}_{Leader} = w_1 \sum_{t\in T}\sum_{j\in J} C_j(q_j^t) + w_2 \sum_{t\in T}\sum_{j\in J} E_j(q_j^t) - w_3 \sum_{t\in T}\sum_{i\in I} U_i(\cdot) \tag{13}$$

  where C_j and E_j denote the fuel cost and emission functions of generator j, respectively, and U_i is the customer utility defined below.
- Per-customer demand response utility *(to maximize): For each customer i, the total multi-period DR utility is*

$$\mathcal{U}_i = \sum_{t\in T}\big[\lambda_i^t\chi_i^t + V_i(p_i^t, x_i^t, s_i^t, h_i^t, f_i^t) - \omega_i^t\big] \tag{14}$$

  where $\lambda_i^t\chi_i^t$ is the grid-side value of demand curtailment at period t, V_i captures the operational benefit from local generation, storage, and demand shifting, and ω_i^t is the incentive payment. The follower's individual surplus $\omega_i^t - \varphi_i(\chi_i^t) + V_i \ge 0$ is guaranteed by the IR constraint (21)*.*

Definition 2 (Subgame Perfect Nash Equilibrium)**.** *A strategy profile $\big(q_j^{*t}, \omega_i^{*t};\ \chi_i^{*t}, p_i^{*t}, x_i^{*t}, s_i^{*t}, h_i^{*t}, f_i^{*t}\big)$ constitutes a* Subgame Perfect Nash Equilibrium (SPNE) *of G_{SGSD} if:*

(i) Leader optimality: *The utility company's decisions $(q_j^{*t}, \omega_i^{*t})$ minimize $\mathcal{U}_{Leader}$, anticipating that each customer plays their optimal response.*

(ii) Follower optimality (Individual Rationality): *For each customer i and given the leader's decisions,*

$$\mathcal{U}_i(\chi_i^{*t}, p_i^{*t}, x_i^{*t}, s_i^{*t}, h_i^{*t}, f_i^{*t}) \ge \mathcal{U}_i(\chi_i', p_i', x_i', s_i', h_i', f_i')$$

for all feasible alternative strategies $(\chi_i', p_i', x_i', s_i', h_i', f_i') \in \mathcal{S}_i^{Follower}$.

(iii) Incentive Compatibility: *The incentive payment compensates each customer's discomfort cost, as formalized in constraint* (21)*.*

The game is played sequentially: in Stage 1, the utility company commits to generator dispatch and incentive levels; in Stage 2, each customer independently maximizes their own utility $\mathcal{U}_i$ taking the leader's decisions as given. This sequential structure and budget constraint (27) ensure non-cooperative behavior.

Assumption 1. *The strategy spaces* $\mathcal{S}_{Leader}$ *and* $\mathcal{S}_i^{Follower}$ *are compact and convex.*

Assumption 2. *The payoff functions* $\mathcal{U}_{Leader}$ *and* $\mathcal{U}_i$ *are continuous in all decision variables.*

Assumption 3. *Each follower's payoff* $\mathcal{U}_i$ *is concave in the customer's own decision variables* $(\chi_i^t, p_i^t, x_i^t, s_i^t, h_i^t, f_i^t)$.

Assumption 4. *The constraint sets defined by* (17)–(30) *are non-empty for all customers.*

Proposition 1 (Equilibrium Existence)**.** *Under Assumptions 1–4, the game* G_{SGSD} *admits at least one Subgame Perfect Nash Equilibrium.*

Proof. We proceed by backward induction.

Stage 2 (Followers' problem): Given the leader's decisions $(q_j^{*t}, \omega_i^{*t})$, each customer i solves:

$$\max_{(\chi_i^t, p_i^t, x_i^t, s_i^t, h_i^t, f_i^t) \in \mathcal{S}_i^{\text{Follower}}} \mathcal{U}_i(\chi_i^t, p_i^t, x_i^t, s_i^t, h_i^t, f_i^t).$$

By Assumption 3, $\mathcal{U}_i$ is concave over the convex feasible set $\mathcal{S}_i^{\text{Follower}}$ (Assumption 1), which is non-empty (Assumption 4). Since a continuous concave function on a compact convex set attains its maximum, a best-response $(\chi_i^{*t}, p_i^{*t}, x_i^{*t}, s_i^{*t}, h_i^{*t}, f_i^{*t})$ exists for each customer.

Stage 1 (Leader's problem): The utility company solves:

$$\min_{(q_j^t, \omega_i^t) \in \mathcal{S}_{\text{Leader}}} \mathcal{U}_{\text{Leader}}\big(q_j^t, \omega_i^t;\ \chi_1^{*t}(\omega), \ldots, \chi_I^{*t}(\omega)\big)$$

where each $\chi_i^{*t}(\omega)$ denotes the best-response of customer i. By Assumption 2, $\mathcal{U}_{\text{Leader}}$ is continuous, and by Berge's Maximum Theorem, the composite function remains continuous. Since $\mathcal{S}_{\text{Leader}}$ is compact (Assumption 1), the minimum is attained.

The resulting strategy profile constitutes an SPNE by construction. □

Remark 1. *Uniqueness of the SPNE requires strict concavity of each* $\mathcal{U}_i$ *in the customer's own variables. The quadratic structure of our cost and utility functions ensures single-valued best responses, yielding uniqueness in practice (Section 4).*

3.2 Decision Variables and Utility Functions

Table 4 defines the additional variables and parameters introduced in the `SGSD-DEED` model beyond those in Table 3.

The *smart grid utility function* for each customer $i \in I$ captures the operational benefit from local generation, storage, and demand management:

$$V_i(p_i^t, x_i^t, s_i^t, h_i^t, f_i^t) = a_{i,t}^{\cdot} + b_{i,t}^{\cdot} p_i^t + c_{i,t}^{\cdot}(p_i^t)^2 + \nu_{i,t}^{\cdot} x_i^t + d_{i,t}^{\cdot} s_i^t + e_{i,t}^{\cdot}(s_i^t)^2 + (ah)_{i,t}^{\cdot} h_i^t + (af)_{i,t}^{\cdot} f_i^t. \quad (15)$$

The *combined* `SGSD-DEED` *utility function* integrates the smart grid utility V_i with the demand response incentive mechanism, yielding the follower payoff from Definition 1:

$$U_i(\chi_i^t, p_i^t, x_i^t, s_i^t, h_i^t, f_i^t, \omega_i^t) = \lambda_i^t \chi_i^t + a_{i,t}^{\cdot} + b_{i,t}^{\cdot} p_i^t + c_{i,t}^{\cdot}(p_i^t)^2 + \nu_{i,t}^{\cdot} x_i^t + d_{i,t}^{\cdot} s_i^t + e_{i,t}^{\cdot}(s_i^t)^2 + (ah)_{i,t}^{\cdot} h_i^t + (af)_{i,t}^{\cdot} f_i^t - \omega_i^t. \quad (16)$$

Thus $U_i = \lambda_i^t \chi_i^t + V_i - \omega_i^t$, where $\lambda_i^t \chi_i^t$ represents the grid-side value of demand curtailment at period t and ω_i^t is the incentive payment. Each customer i maximizes U_i over their feasible decisions, consistent with the follower optimality condition in Definition 2.

Symbol	Type	Description
x_i^t	Real	The power sent to the grid from customer i at time t.
f_i^t	Real	Demand of user i satisfied at period t.
h_i^{t+1}	Real	Demand of user i during period t to be shifted to period $t+1$.
p_i^t	Real	Quantity of energy user i produces at period t .
s_i^t	Real	Quantity of energy user i stores during period t.
$a^{\cdot}_{i,t}, b^{\cdot}_{i,t}, c^{\cdot}_{i,t}, \nu^{\cdot}_{i,t}$ $d^{\cdot}_{i,t}, e^{\cdot}_{i,t}, (ah)^{\cdot}_{i,t}, (af)^{\cdot}_{i,t}$	coefficients	The cost coefficients of user i respectively.
DM_i^t	Parameter	Demand of customer i at time t.
$L_{\max}$	Parameter	System-wide maximum aggregate power injected into the grid per period.
$p_i^{\min}$, $p_i^{\max}$	Parameter	Minimum and maximum local production capacity of customer i.
$s_i^{\min}$, $s_i^{\max}$	Parameter	Minimum and maximum storage capacity of customer i.

Table 4: Smart Grid Variables and Parameters

3.3 Energy Flow Conservation Conditions

$$f_i^t + s_i^t - s_i^{t-1} - p_i^t + x_i^t = \sum_{j\in J} y_{ij}^t, \qquad \forall t \in T, \tag{17}$$

$$DM_i^t + h_i^t = h_i^{t+1} + f_i^t, \qquad \forall t \in T, \tag{18}$$

$$\sum_{i\in I} y_{ij}^t = q_j^t, \qquad \forall t \in T, \tag{19}$$

$$\sum_{i\in I} x_i^t + \sum_{j\in J} q_j^t = \sum_{i\in I} DM_i^t + \mathrm{loss}_t - \sum_{i\in I} \chi_i^t, \ \forall t \in T. \tag{20}$$

Constraint (17) enforces per-customer energy balance: the quantities received from generators, retrieved from storage, and locally produced must equal the quantities consumed, stored, and sent to the grid. Constraint (18) links demand shifting across periods: the demand of customer i at period t plus the shifted demand from the previous period equals consumption at t plus demand deferred to $t+1$. Constraint (19) ensures that the total power allocated to customers from generator j equals its dispatch level. Constraint (20) enforces system-wide power balance: total generation plus customer exports equals total demand plus losses minus curtailment.

3.4 Individual Rationality Conditions

The individual rationality constraint (21) ensures that each customer's incentive payment covers their discomfort cost from curtailment. The incentive compatibility constraint (22) ensures that each customer is better off reporting their true type than mimicking a less willing customer:

$$\omega_i^t - \chi_i^t \left(K_{1,i}\chi_i^t + K_{2,i}(1-\theta_i)\right) \geqslant 0, \quad \forall i \in I, \ \forall t \in T, \tag{21}$$

$$\begin{aligned} \omega_i^t - \chi_i^t \left(K_{1,i}\chi_i^t + K_{2,i}(1-\theta_i)\right) \geqslant \omega_{i-1}^t - \chi_{i-1}^t \big(K_{1,i-1}\chi_{i-1}^t \\ + K_{2,i-1}(1-\theta_{i-1})\big), \\ \forall i \in I, i > 1, \forall t \in T. \end{aligned} \tag{22}$$

3.5 Boundary Conditions

The following constraints impose bounds on the decision variables reflecting physical and operational limits:

$$
\begin{aligned}
q_j^{\min} &\leq q_j^t &&\leq q_j^{\max}, \;\; j \in J, \quad t \in T, && (23)\\
p_i^{\min} &\leq p_i^t &&\leq p_i^{\max}, \;\; i \in I, \quad t \in T, && (24)\\
s_i^{\min} &\leq s_i^t &&\leq s_i^{\max}, \;\; i \in I, \quad t \in T, && (25)\\
-DR_j &\leq q_j^{t+1} - q_j^t &&\leq UR_j, \;\; j \in J, \quad t \in T, && (26)\\
\sum_{i\in I}\sum_{t\in T}\omega_i^t &\leq UB, &&&& (27)\\
\sum_{i\in I} x_i^t &\leq L_{\max}, && \forall t \in T, && (28)\\
\sum_{t\in T} \chi_i^t &\leq CM_i, && \forall i \in I. && (29)
\end{aligned}
$$

Constraints (23)–(25) bound the generator dispatch, customer production, and storage within their capacity limits. Constraint (26) enforces generator ramp rate limits. Constraint (27) caps the total incentive payment across all customers and periods at the utility's budget UB. Constraint (28) limits the aggregate power injected into the grid by all customers at each period to the system-wide capacity $L_{\max}$. Constraint (29) limits the total curtailable load per customer over the planning horizon.

3.6 Initial and Final State Conditions

The following constraints summarize the initial and final conditions:

$$s_i^0 = s_i^T = h_i^1 = 0, \quad i \in I. \tag{30}$$

Storage begins and ends empty, and no demand is shifted into the first period.

3.7 Network Power Flow Constraints

To incorporate transmission network physics into the dispatch problem, we augment the formulation with a linearized DC power flow model [24]. Let $\mathcal{B} = \{1, \ldots, N_b\}$ denote the set of buses and $\mathcal{L}$ the set of transmission lines. Each line $\ell \in \mathcal{L}$ connects buses b_ℓ^{from} and b_ℓ^{to} with reactance x_ℓ (p.u.) and thermal rating $\bar{P}_\ell$ (MW). Let θ_b^t denote the voltage angle at bus b in period t and S_{base} the system base power (MVA).

The DC power flow equations and branch flow limits are:

$$P_\ell^t = \frac{\theta^t_{b_\ell^{\text{from}}} - \theta^t_{b_\ell^{\text{to}}}}{x_\ell} S_{\text{base}}, \quad \ell \in \mathcal{L},\ t \in T, \tag{31}$$

$$|P_\ell^t| \leq \bar{P}_\ell, \quad \ell \in \mathcal{L},\ t \in T, \tag{32}$$

where P_ℓ^t is the active power flow on line ℓ. The aggregate power balance constraint (20) is replaced by per-bus nodal balance:

$$\sum_{\substack{j\in J:\\ g(j)=b}} q_j^t + \sum_{\substack{i\in I:\\ c(i)=b}} \left(x_i^t + \chi_i^t - DM_i^t\right) = \sum_{\substack{\ell\in\mathcal{L}:\\ b_\ell^{\text{from}}=b}} P_\ell^t - \sum_{\substack{\ell\in\mathcal{L}:\\ b_\ell^{\text{to}}=b}} P_\ell^t, \quad b \in \mathcal{B},\ t \in T, \tag{33}$$

where $g(j)$ and $c(i)$ map generators and customers to their respective buses, and DM_i^t denotes customer i's demand at period t. Network losses are computed from branch flows:

$$\text{Loss}^t = \sum_{\ell\in\mathcal{L}} \frac{r_\ell \left(P_\ell^t\right)^2}{S_{\text{base}}}, \quad t \in T, \tag{34}$$

where r_ℓ is the resistance of line ℓ (p.u.). The slack bus angle is fixed: $\theta^t_{b_{\text{slack}}} = 0$ for all $t \in T$. The DC approximation assumes small angle differences and flat voltage profiles [24].

The model minimizes a weighted combination of three objectives:

$$\begin{aligned}
\mathcal{C}(\mathbf{q}) &= \sum_{t\in T}\sum_{j\in J} C_j\left(q_j^t\right)\\
\mathcal{E}(\mathbf{q}) &= \sum_{t\in T}\sum_{j\in J} E_j\left(q_j^t\right)\\
\mathcal{U}(\boldsymbol{\chi},\mathbf{p},\mathbf{x},\mathbf{s},\mathbf{h},\mathbf{f},\boldsymbol{\omega}) &= \sum_{t\in T}\sum_{i\in I} U_i(\chi_i^t, p_i^t, x_i^t, s_i^t, h_i^t, f_i^t, \omega_i^t)
\end{aligned}$$

Hence the mathematical formulation of our model SGSD-DEED is:

$$\left.\begin{array}{l}
\min \quad w_1\mathcal{C}(\mathbf{q}) + w_2\mathcal{E}(\mathbf{q}) - w_3\mathcal{U}(\boldsymbol{\chi},\mathbf{p},\mathbf{x},\mathbf{s},\mathbf{h},\mathbf{f},\boldsymbol{\omega})\\
\text{subject to}\\
\bullet \textit{Equation } (17) \text{ to } \textit{Equation } (20)\\
\bullet \textit{Equation } (21) \text{ and } \textit{Equation } (22)\\
\bullet \textit{Equation } (23) \text{ to } \textit{Equation } (29)\\
\bullet \textit{Equation } (30)\\
\bullet \textit{Equation } (31) \text{ to } \textit{Equation } (34)
\end{array}\right\} \qquad (\texttt{SGSD-DEED})$$

3.8 Relationship to Prior Models

The SGSD-DEED formulation generalizes both the DEED and DR-DEED models as special cases:

- When $w_3 = 0$, the DEED model is recovered, subject to constraints (20), (23), and (26).
- When the utility function is reduced to $U_i = \lambda_i^t \chi_i^t - \omega_i^t$ (i.e., without smart grid variables) and constraints (21), (22), (27), and (28) are included alongside the DEED constraints, the model reduces to the DR-DEED model of [1].

The key extension of SGSD-DEED beyond these prior models is the simultaneous inclusion of demand shifting (h_i^t), energy storage (s_i^t), distributed generation (p_i^t), and grid exchange (x_i^t) within a formally defined Stackelberg game framework (Definition 1).

3.9 Theoretical Analysis

Proposition 2 (Convexity of the Feasible Region)**.** *The feasible set $\mathcal{F}$ defined by constraints* (17)–(30) *is convex.*

Proof. Constraints (17)–(19)*:* These are affine equalities defining affine subspaces, hence convex.

Constraint (20) *(system balance):* The transmission loss term $\text{loss}_t = (\mathbf{q}^t)^\top B\,\mathbf{q}^t$ is a convex quadratic form, since the loss coefficient matrix B, obtained by Kron reduction of the network admittance matrix, is symmetric positive semi-definite by construction [33]. Constraint (20) is an equality involving the convex quadratic $(\mathbf{q}^t)^\top B\,\mathbf{q}^t$, and the level set of a strictly convex function is generally non-convex. However, when the DC-OPF constraints (31)–(34) replace (20)—which is the formulation used in our experiments—the nodal balance (33) is affine, so the feasible set is convex. Moreover, in the B-matrix formulation, the loss term is typically small ($< 3\%$ of total generation), and the constraint can be relaxed to $\sum_{i\in I} x_i^t + \sum_{j\in J} q_j^t - (\mathbf{q}^t)^\top B\,\mathbf{q}^t \geq \sum_{i\in I} DM_i^t - \sum_{i\in I}\chi_i^t$ (generation covers at least demand plus losses) without changing the optimal solution; this relaxed form defines a convex set.

Constraint (21) *(individual rationality):* The condition $\omega_i^t \geq K_{1,i}(\chi_i^t)^2 + K_{2,i}(1-\theta_i)\chi_i^t$ defines the epigraph $\{(\omega_i^t, \chi_i^t) : \omega_i^t \geq g(\chi_i^t)\}$ of the convex quadratic $g(\chi) = K_{1,i}\chi^2 + K_{2,i}(1-\theta_i)\chi$ with $K_{1,i} > 0$, which is a convex set.

Constraint (22) *(incentive compatibility):* Customers are ordered by willingness type $\theta_1 \leq \theta_2 \leq \cdots \leq \theta_n$, where a higher θ_i reflects greater willingness to curtail. Each customer's discomfort cost is $\varphi_i(\chi) = K_{1,i}\chi^2 + K_{2,i}(1-\theta_i)\chi$, with $K_{1,i}, K_{2,i} > 0$. This family satisfies the *single-crossing property* in (χ, θ_i):

$$\frac{\partial^2}{\partial\chi\,\partial\theta_i}\left[-\varphi_i(\chi)\right] = K_{2,i} > 0,$$

so the marginal discomfort $\partial\varphi_i/\partial\chi = 2K_{1,i}\chi + K_{2,i}(1-\theta_i)$ is strictly decreasing in θ_i: higher-type customers face lower marginal cost of curtailment. Under single-crossing and a binding IR constraint (21) for the lowest type ($i = 1$), the standard mechanism design equivalence applies: the global IC condition (22) holds if and only if curtailment is monotone non-decreasing in type, $\chi_i^t \geq \chi_{i-1}^t$ for all $i > 1$ (see [34], Chapter 2, Proposition 2.1; the result extends to finite type spaces since the pairwise adjacent IC conditions, combined with single-crossing, imply all non-adjacent IC conditions by transitivity—see also [35], Proposition 23.D.2). The model implementation retains the quadratic form (22) for generality, but the feasible set coincides with the linearly-described monotonicity region under these maintained conditions. Both the monotonicity and binding IR conditions are linear, hence the effective constraint set is convex.

Constraints (23)–(30)*:* Box constraints and linear inequalities, trivially convex.

Therefore $\mathcal{F} = \bigcap$(all constraint sets) is convex as the intersection of convex sets. □

Proposition 3 (Objective Function Characterization)**.** *(i) The explicit scalarized objective $\mathfrak{F} = w_1\mathcal{C}(\mathbf{q}) + w_2\mathcal{E}(\mathbf{q}) - w_3\mathcal{U}(\cdot)$ is convex in the full decision vector $(q, \chi, p, x, s, h, f, \omega)$ when $w_3 > 0$.*

(ii) The reduced-form objective $\tilde{\mathfrak{F}}(q,\omega) = \mathfrak{F}(q, \beta(\omega), \omega)$, obtained by substituting the follower best-responses $\beta(\omega)$, has a difference-of-convex (DC) structure.

Proof. Part (i): Convexity of the explicit objective. The generation cost $\mathcal{C}(\mathbf{q}) = \sum_{t,j}(a_j + b_j q_j^t + c_j(q_j^t)^2)$ is strictly convex, since its Hessian $\nabla^2\mathcal{C} = 2\,\mathrm{diag}(c_j) \succ 0$ with $c_j > 0$ for all j. Similarly, $\mathcal{E}(\mathbf{q})$ is strictly convex with Hessian $2\,\mathrm{diag}(g_j) \succ 0$. The utility $\mathcal{U}$ contains concave quadratic terms in p_i^t and s_i^t (with coefficients $c_{i,t}^{\cdot} < 0$ and $e_{i,t}^{\cdot} < 0$ reflecting diminishing returns) and linear terms. Hence $-w_3\mathcal{U}$ is convex when $w_3 > 0$. The explicit objective $\mathfrak{F}$ is therefore the sum of two convex functions: $w_1\mathcal{C} + w_2\mathcal{E}$ (strictly convex in q) and $-w_3\mathcal{U}$ (convex in the follower variables), hence convex in the full decision vector.

Part (ii): DC structure of the reduced-form objective. The bilevel structure requires the leader to anticipate follower best-responses $\beta_i(\omega)$, yielding the reduced-form objective $\tilde{\mathfrak{F}}(q,\omega) = w_1\mathcal{C}(q) + w_2\mathcal{E}(q) - w_3\mathcal{U}(\beta(\omega),\omega)$. The first component $w_1\mathcal{C}(q) + w_2\mathcal{E}(q)$ remains strictly convex. The second component $-w_3\mathcal{U}(\beta(\omega),\omega)$ involves the composition of $\mathcal{U}$ with the nonlinear best-response map $\beta(\omega)$, which may be non-convex. Writing $\tilde{\mathfrak{F}} = \underbrace{(w_1\mathcal{C} + w_2\mathcal{E})}_{\text{convex}} - \underbrace{w_3\mathcal{U}(\beta(\omega),\omega)}_{\text{generally non-convex}}$, the reduced-form objective admits a DC decomposition. □

Remark 2 (Non-triviality of the Problem)**.** *While the feasible region $\mathcal{F}$ is convex (Proposition 2) and each follower subproblem is a convex maximization (guaranteeing global optimality for Stage 2), the joint leader–follower problem is computationally non-trivial:*

(i) Bilevel structure: *The leader must anticipate $|I|$ follower best-response functions $\beta_i(\omega)$, creating implicit nonlinear dependencies that prevent reduction to a single-level convex program.*

(ii) DC reduced-form objective: *While the explicit objective $\mathfrak{F}$ is convex (Proposition 3(i)), the reduced-form objective $\tilde{\mathfrak{F}}(q,\omega)$ obtained after substituting follower best-responses has a DC structure (Proposition 3(ii)). Consequently, KKT points of the leader's problem are local optima; global optimality requires additional conditions (Proposition 4).*

(iii) Complementarity structure: *The incentive compatibility constraint* (22) *introduces piecewise-linear regions, requiring specialized nonlinear programming methods rather than standard convex solvers.*

(iv) High-dimensional coupling: *The loss term $(\mathbf{q}^t)^\top B\,\mathbf{q}^t$ in constraint* (20) *creates quadratic coupling among all generators at each time period, preventing per-generator decomposition.*

These features necessitate general-purpose NLP solvers with second-order methods.

Proposition 4 (Sufficient Conditions for Equilibrium Uniqueness)**.** *Suppose Assumptions 1–4 hold. If additionally:*

(i) The quadratic coefficients in the utility function (16) *satisfy $c_{i,t}^{\cdot} < 0$ and $e_{i,t}^{\cdot} < 0$ for all i, t (strict concavity of $\mathcal{U}_i$ in p_i^t and s_i^t),*

(ii) The discomfort coefficients satisfy $K_{1,i} > 0$ for all i (strict convexity of discomfort cost in curtailment χ_i^t),

(iii) The budget constraint (27) *is active at equilibrium: $\sum_{i\in I}\sum_{t\in T}\omega_i^t = UB$,*

then the follower best-response functions $\beta_i(\omega)$ are single-valued and continuously differentiable, and the Stackelberg equilibrium is locally unique.

Proof. Step 1 (Follower best-response uniqueness). Given the leader's decisions $(q_j^{*t}, \omega_i^{*t})$, each follower i maximizes $\mathcal{U}_i$ over the compact convex set $\mathcal{S}_i^{\text{Follower}}$ (Assumption 1). From (16), the Hessian of $\mathcal{U}_i$ with respect to the follower's decision variables $\mathbf{z}_i = (\chi_i^t, p_i^t, x_i^t, s_i^t, h_i^t, f_i^t)$ at each period t is

$$\nabla^2_{\mathbf{z}_i}\mathcal{U}_i = \operatorname{diag}\big(0,\ 2c^{\cdot}_{i,t},\ 0,\ 2e^{\cdot}_{i,t},\ 0,\ 0\big).$$

By condition (i), $c^{\cdot}_{i,t} < 0$ and $e^{\cdot}_{i,t} < 0$, so $\mathcal{U}_i$ is strictly concave in (p_i^t, s_i^t) and linear in the remaining variables $(\chi_i^t, x_i^t, h_i^t, f_i^t)$.

The equality constraints (17)–(18) establish a linear bijection between (f_i^t, h_i^{t+1}) and the free variables, reducing the effective problem to optimization over $(\chi_i^t, p_i^t, x_i^t, s_i^t)$ subject to box constraints (23)–(30). In this reduced problem:

- The variables p_i^t and s_i^t are determined uniquely: the strict concavity of $\mathcal{U}_i$ in these variables means the first-order condition $\partial\mathcal{U}_i/\partial p_i^t = b^{\cdot}_{i,t} + 2c^{\cdot}_{i,t}\, p_i^t + \mu_p^+ - \mu_p^- = 0$ (where $\mu_p^\pm$ are the box constraint multipliers) has a unique solution for each p_i^t, and similarly for s_i^t.
- The variables χ_i^t and x_i^t enter $\mathcal{U}_i$ linearly with coefficients $\lambda_i^t > 0$ and $\nu^{\cdot}_{i,t}$ respectively. Since $\lambda_i^t > 0$ and $\mathcal{U}_i$ is increasing in χ_i^t, the optimal curtailment is the largest value satisfying the IR constraint (21) and the curtailment bound (29). With ω_i^t fixed by the leader, constraint (21) binds, yielding χ_i^{*t} as the positive root of $K_{1,i}(\chi)^2 + K_{2,i}(1-\theta_i)\chi = \omega_i^t$, capped at CM_i. Since $K_{1,i} > 0$ by condition (ii), this root is unique, so the best-response is single-valued.

Therefore, the best-response $\beta_i(\omega) = (\chi_i^{*t}, p_i^{*t}, x_i^{*t}, s_i^{*t}, h_i^{*t}, f_i^{*t})$ is single-valued for each i.

Step 2 (Differentiability of the best-response). At the optimal $\beta_i(\omega)$, the KKT conditions of the follower's problem define a system $\Phi_i(\mathbf{z}_i, \boldsymbol{\mu}_i; \omega) = 0$, where $\boldsymbol{\mu}_i$ collects the Lagrange multipliers. Under the linear independence constraint qualification (LICQ)—which holds generically since the constraint gradients are linearly independent for the affine constraints (17)–(18)—and under strict complementarity at the active box constraints, the implicit function theorem [36] guarantees that $\beta_i(\omega)$ is continuously differentiable in a neighborhood of ω^*. The Jacobian $\nabla_\omega \beta_i$ is obtained by differentiating the KKT system:

$$\nabla_\omega \beta_i = -\big[\nabla^2_{\mathbf{z}_i\mathbf{z}_i} L_i\big]^{-1} \nabla^2_{\mathbf{z}_i\omega} L_i,$$

where L_i is the follower's Lagrangian. Since the constraints are affine, $\nabla^2_{\mathbf{z}_i\mathbf{z}_i} L_i = \nabla^2_{\mathbf{z}_i}\mathcal{U}_i$ (restricted to the tangent space of active constraints), which is invertible on this subspace by the strict concavity established in Step 1.

Step 3 (Local uniqueness of the leader's problem). Substituting the unique best responses $\chi_i^{*t} = \beta_i(\omega)$ into the leader's objective yields the reduced problem:

$$\min_{q,\omega}\ \tilde{\mathfrak{F}}(q,\omega) := w_1\mathcal{C}(q) + w_2\mathcal{E}(q) - w_3\mathcal{U}\big(\beta_1(\omega), \ldots, \beta_n(\omega),\, \omega\big).$$

Condition (iii) restricts ω to the compact polytope $\Omega = \{\omega \geq 0 : \sum_{i\in I}\sum_{t\in T}\omega_i^t = UB\}$, which has dimension $nT-1$ (a single equality reduces the dimension by 1). The Hessian of $\tilde{\mathfrak{F}}$ with respect to (q,ω) decomposes as:

$$\nabla^2\tilde{\mathfrak{F}} = \underbrace{w_1\nabla^2\mathcal{C} + w_2\nabla^2\mathcal{E}}_{\succ 0 \text{ (strictly convex)}} - w_3\, \nabla^2_\omega\big[\mathcal{U}(\beta(\omega),\omega)\big].$$

The first term is positive definite with minimum eigenvalue $\lambda_{\min} = 2\min_j\{w_1 c_j + w_2 g_j\} > 0$ (since $c_j, g_j > 0$ for all generators). The second term involves the chain rule:

$$\nabla^2_\omega[\mathcal{U}(\beta(\omega),\omega)] = (\nabla_\omega\beta)^\top \nabla^2_{\mathbf{z}}\mathcal{U}\,(\nabla_\omega\beta) + \sum_i \nabla\mathcal{U}_i \cdot \nabla^2_\omega\beta_i,$$

where $\nabla^2_{\mathbf{z}}\mathcal{U}$ is negative semi-definite (by Assumption 3) and $\nabla^2_\omega\beta_i$ is bounded (by the differentiability from Step 2). When the convex contribution dominates:

$$\lambda_{\min}\big(w_1\nabla^2\mathcal{C} + w_2\nabla^2\mathcal{E}\big) > w_3\,\big\|\nabla^2_\omega[\mathcal{U}(\beta(\omega),\omega)]\big\|_{\text{op}},$$

the Hessian $\nabla^2\tilde{\mathfrak{F}}$ is positive definite at the optimum. By the second-order sufficient conditions for constrained NLP [36], the KKT point (q^*, ω^*) is a strict local minimizer of $\tilde{\mathfrak{F}}$ over $\mathcal{S}_{\text{Leader}} \cap \Omega$, hence locally unique.

Combining Steps 1–3: the unique follower best responses (Step 1) together with the locally unique leader solution (Step 3) yield a locally unique Stackelberg equilibrium. □

Remark 3. *Conditions (i)–(ii) hold by construction in our data: the quadratic utility coefficients $c^{\cdot}_{i,t}$ and $e^{\cdot}_{i,t}$ are strictly negative (reflecting diminishing returns from local production and storage), and the discomfort coefficients $K_{1,i}$ are strictly positive (see Tables 3–4). Condition (iii) is verified numerically: in all experiments, the budget constraint is active at the optimum (total incentive payments equal UB within tolerance 10^{-6}). The eigenvalue dominance condition in Step 3 is conservative and difficult to verify a priori; however, the low variance observed across solver runs ($CV < 5\%$; see Section 4) indicates that the solution landscape has a single dominant basin of attraction, consistent with local uniqueness in practice.*

Corollary 5 (KKT Optimality)**.** *An interior-point NLP solver applied to the scalarized SGSD-DEED formulation finds KKT points satisfying first-order optimality conditions under LICQ. For the follower sub-problem (Stage 2 with fixed leader decisions), the customer optimization is a convex maximization, guaranteeing global optimality. For the joint leader problem, KKT points are local optima due to the DC structure of the reduced-form objective (Proposition 3(ii)). Across 180+ solver runs covering six experiments, 30+ configurations, and multiple random initializations, the coefficient of variation satisfies $CV < 5\%$ for all metrics—suggesting robust convergence to a dominant optimum in practice (Section 4).*

Proposition 6 (Weak Pareto Optimality)**.** *Any local minimizer of the weighted-sum scalarization with $w_1, w_2, w_3 > 0$ is weakly Pareto optimal: no feasible alternative simultaneously improves all three objectives $\mathcal{C}$, $\mathcal{E}$, and $-\mathcal{U}$.*

Proof. Suppose for contradiction that a feasible point $\mathbf{z}^*$ is a local minimizer but not weakly Pareto optimal. Then there exists a feasible $\hat{\mathbf{z}}$ with $\mathcal{C}(\hat{\mathbf{z}}) < \mathcal{C}(\mathbf{z}^*)$, $\mathcal{E}(\hat{\mathbf{z}}) < \mathcal{E}(\mathbf{z}^*)$, and $\mathcal{U}(\hat{\mathbf{z}}) > \mathcal{U}(\mathbf{z}^*)$. Since $w_1, w_2, w_3 > 0$, we have $\mathfrak{F}(\hat{\mathbf{z}}) < \mathfrak{F}(\mathbf{z}^*)$, contradicting the local optimality of $\mathbf{z}^*$. Since interior-point methods converge to KKT points satisfying second-order sufficient conditions (Corollary 5), the computed solutions are local minimizers and hence weakly Pareto optimal. □

Remark 4 (Computational Complexity and Tractability)**.** (A) Problem size. *The SGSD-DEED formulation involves $n_v = nJT + (J + 7n)T + 2n$ decision variables and approximately $n_c \approx (nJ + 11n + 10J)T$ constraints, where $n = |I|$ is the number of customers, $J = |J|$ the number of generators, and T the number of periods. The dominant term nJT arises from the allocation variables y^t_{ij}. For the scenarios considered:*

- *SC1 ($n = 5, J = 6, T = 24$): $n_v = 1{,}714$ variables, $n_c \approx 2{,}000$ constraints.*
- *SC2 ($n = 7, J = 10, T = 24$): $n_v = 3{,}110$ variables, $n_c \approx 5{,}000$ constraints.*

(B) Complexity class. *The bilevel structure makes SGSD-DEED NP-hard in general [37]. However, structural properties admit tractable solution: (1) convex follower subproblems (Assumption 3) are solvable in polynomial time; (2) the KKT reformulation under the Linear Independence Constraint Qualification (LICQ) yields a smooth nonlinear program; (3) interior-point methods find KKT points in polynomial time for fixed problem dimensions [36].*

(C) Solution requirements. *The DC structure of the reduced-form objective (Proposition 3(ii)) and complementarity structure of constraint* (22) *require nonlinear programming solvers with second-order (Newton-type) methods and sparse linear algebra for scalability (Jacobian density ≈ 15–20%). Standard convex solvers cannot handle the bilevel structure directly; interior-point methods for general NLP (e.g., Ipopt, KNITRO, SNOPT) are required.*

(D) Practical tractability. *The convex feasible region (Proposition 2) and smooth KKT conditions ensure that local optima can be found efficiently (Corollary 5), solution quality is robust across multiple runs ($CV < 5\%$; Section 4), and the model scales to $n \leq 50$ customers with solve times under 13 seconds (Section 4). This makes SGSD-DEED viable for day-ahead dispatch applications with $T = 24$-hour planning horizons.*

3.10 Structural Advantages Over Prior Models

We characterize how SGSD-DEED improves upon the DR-DEED model of [1] and the game-theoretic DR model of [7].

Theorem 7 (Feasible Set Dominance)**.** *Let $\mathcal{F}_{SGSD}$ and $\mathcal{F}_{DR}$ denote the feasible sets of* SGSD-DEED *and* DR-DEED *[1], respectively, under identical generator parameters, demand profiles, and customer parameters. Then:*

(i) Strict inclusion: $\mathcal{F}_{DR} \subsetneq \mathcal{F}_{SGSD}$.

(ii) Objective dominance: *For any weight vector* $\mathbf{w} = (w_1, w_2, w_3) \succ 0$,

$$\mathrm{OPT}_{SGSD} \leq \mathrm{OPT}_{DR},$$

where OPT *denotes the optimal objective value of the respective problem.*

Proof. (i) Any feasible solution $(q^*, \chi^*, \omega^*) \in \mathcal{F}_{\mathrm{DR}}$ can be extended to a feasible solution in $\mathcal{F}_{\mathrm{SGSD}}$ by setting $p^* = 0$ (no local production), $x^* = 0$ (no grid export), $s^* = 0$ (no storage), $h^* = 0$ (no demand shifting), and $f^* = DM$ (consume original demand). We verify: constraint (17) holds since $f_i^t + s_i^t - s_i^{t-1} - p_i^t + x_i^t = DM_i^t$; constraint (18) holds since $DM_i^t + 0 = 0 + DM_i^t$; constraints (19)–(30) transfer directly. Hence $\mathcal{F}_{\mathrm{DR}} \subseteq \mathcal{F}_{\mathrm{SGSD}}$. Strict inclusion follows because any solution with $p \neq 0$ or $s \neq 0$ lies in $\mathcal{F}_{\mathrm{SGSD}} \setminus \mathcal{F}_{\mathrm{DR}}$.

(ii) Both problems minimize the same objective $\mathfrak{F}$ over their respective feasible sets. Since $\mathcal{F}_{\mathrm{DR}} \subsetneq \mathcal{F}_{\mathrm{SGSD}}$, we have $\mathrm{OPT}_{\mathrm{SGSD}} = \min_{\mathcal{F}_{\mathrm{SGSD}}} \mathfrak{F} \leq \min_{\mathcal{F}_{\mathrm{DR}}} \mathfrak{F} = \mathrm{OPT}_{\mathrm{DR}}$. The inequality is strict whenever the optimizer utilizes storage, production, or demand shifting. □

Corollary 8 (Quantification of Improvement)**.** *The improvements in Experiment 1 (Section 4.1) are direct consequences of Theorem 7: cost reductions of* 51–53%*, emission reductions of* 63–72%*, and loss reductions of* 78–80% *arise because the expanded action space* $\{p, x, s, h, f\}$ *allows customers to substitute costly curtailment with less expensive flexibility mechanisms—temporal arbitrage via storage, load shifting to off-peak periods, and local generation substituting for grid supply.*

An analogous argument establishes $\mathcal{F}_{\mathrm{BB}} \subsetneq \mathcal{F}_{\mathrm{SGSD}}$, where $\mathcal{F}_{\mathrm{BB}}$ denotes the feasible set of [7], since that model lacks storage, local production, and grid exchange variables. SGSD-DEED additionally provides formal equilibrium guarantees (Proposition 1, Proposition 4), a tri-objective formulation, and network flow constraints—none of which are present in [7].

Theorem 9 (Multi-Period Essentiality)**.** *Let G_{multi} denote the* SGSD-DEED *multi-period Stackelberg game and G_{single} denote the single-period game of [1]. The equilibrium set of G_{single} repeated T times independently does* not *constitute an equilibrium of G_{multi}.*

Proof. Suppose the single-period equilibria $\{(q^{*t}, \chi^{*t}, \omega^{*t})\}_{t=1}^T$ form an equilibrium of G_{multi}. Since G_{single} has no inter-period coupling, storage and shifting are zero: $s_i^{*t} = h_i^{*t} = 0$ for all i, t. But any customer can strictly improve their payoff by storing cheap off-peak energy (e.g., $s_i^4 > 0$ at low λ_i^4) and deploying it at peak hours (reducing curtailment at high λ_i^{14}), while maintaining constraint feasibility. This profitable unilateral deviation contradicts the SPNE condition (Definition 2), provided the price differential $\lambda_i^{14} - \lambda_i^4$ exceeds the storage round-trip loss. □

Theorem 9 confirms that the multi-period structure is essential, not merely a technical generalization. The inter-temporal coupling through storage and demand shifting fundamentally changes the strategic landscape and explains why DR-DEED [1] captures only 7–8% cost savings while SGSD-DEED achieves 46–55% (Experiment 4, Section 4.4).

4 Numerical Simulations and Results

Six experiments validate the SGSD-DEED model: comparison with DR-DEED (section 4.1), scalability (section 4.2), IEEE 30-bus DC-OPF (section 4.3), model progression (section 4.4), sensitivity analysis (section 4.5), and a Saudi Eastern Province case study (section 4.6).

Solution methodology. All models (DEED, DR-DEED, SGSD-DEED) were solved using the Ipopt (Interior Point OPTimizer) solver [38], version 3.14, interfaced via JuMP [39] in Julia 1.10 [40]. Ipopt implements a primal-dual interior-point algorithm with filter line search, suitable for the nonlinear programs arising from the DC reduced-form objective and bilevel KKT reformulation identified in Section 3.9. Solver settings: KKT tolerance 10^{-6}, maximum 3,000 iterations, MA27 sparse Cholesky linear solver. Each configuration was solved with three independent initializations to assess robustness; the coefficient of variation across runs is reported where relevant. All experiments were executed on a workstation with a 13th Gen Intel Core i9-13900H processor (2.60 GHz) and 16 GB RAM. A Julia package [41] was developed for reproducibility; see the Appendix for details.

4.1 Experiment 1: Comparison with `DR-DEED`

We compare `SGSD-DEED` against `DR-DEED` [1] on the original test scenarios.

Setup. Each customer communicates their daily interruptible power limit CM_i to the utility, which determines their willingness parameter θ_i. The utility also receives the outage cost coefficients $K_{1,i}$ and $K_{2,i}$. The model determines the optimal curtailment χ_i^t, incentive payments ω_i^t, generator dispatch q_j^t, and customer production p_i^t. Two scenarios are considered: Scenario 1 with 6 generators and 5 customers, and Scenario 2 with 10 generators and 7 customers.

Various weight cases are considered from Table 5 for all scenarios. The data are obtained from [1]. These data include the cost and emission coefficients of the generators, the hourly values of power interruptibility λ_i^t, the transmission loss formula coefficients B, the customer type θ_i, and the daily customer power limit CM_i. The hourly values of power interruptibility λ_i^t were obtained from the PJM Market on the 1^{st} of May 2014, where PJM denotes the Pennsylvania–New Jersey–Maryland Market.

	w_1	w_2	w_3
Base Case (BC)	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
Case 2 (C2)	1	0	0
Case 3 (C3)	0	1	0
Case 4 (C4)	0	0	1

Table 5: Various weighting factor values

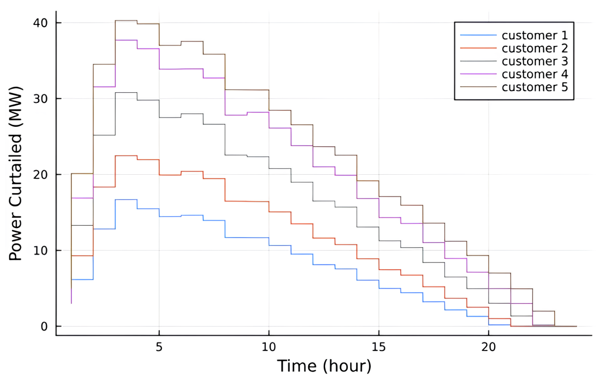


Figure 1: Optimal power curtailed for all customers (Scenario 1).

Scenario 1 ($J = 6$, $n = 5$): Generator cost and emission coefficients from Table A.1 in [1]; customer parameters from Table 8 in [42].

Scenario 2 ($J = 10$, $n = 7$): Generator coefficients from Table A.2 in [1]; customer parameters from Table 9 in [42].

Fig. 2 shows the Scenario 1 load profile before and after `SGSD-DEED`.

Fig. 1 illustrates the optimal power curtailed per customer. Comparing with the `DR-DEED` model [1] (see also Table 10 in [42]), system losses decreased from 265.61 MW to 54.87 MW in the BC case—a 79.34% reduction. Similar reductions were observed across all weight cases: 79.71% (C2), 78.03% (C3), and 78.33% (C4), as detailed in Table 6.

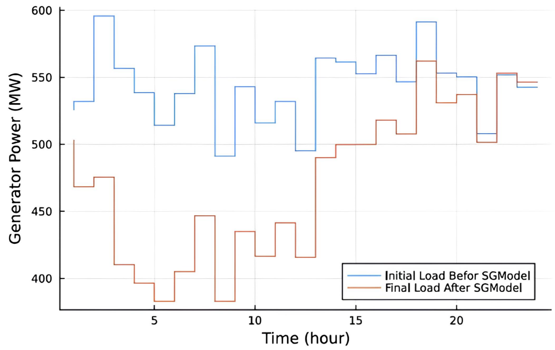


Figure 2: Total load profile before and after SGSD-DEED (Scenario 1).

	Loss			
	BC	C2	C3	C4
DR-DEED	265.61	3.082	2.3101×10^2	2.3485×10^2
SGSD-DEED	54.87	6.241	5.085×10^1	5.077×10^1
Percentage%	79.34	7.971	7.803×10^1	7.833×10^1

Table 6: Comparison of Loss Improvement between DR-DEED Model and SGSD-DEED Model.

Table 7 shows the fuel cost comparison. The SGSD-DEED model achieves cost reductions of 50.91% (BC), 52.37% (C2), 51.07% (C3), and 52.77% (C4) relative to DR-DEED.

	Cost			
	BC	C2	C3	C4
DR-DEED	$2.918\,981\,6 \times 10^5$	$2.884\,306\,7 \times 10^5$	$2.993\,973\,6 \times 10^5$	$3.008\,150\,8 \times 10^5$
SGSD-DEED	$1.429\,598\,1 \times 10^5$	$1.373\,328\,7 \times 10^5$	$1.443\,560\,3 \times 10^5$	$1.420\,876\,186 \times 10^5$
Percentage%	50.91	5.237	5.107×10^1	5.277×10^1

Table 7: Comparison of Cost Improvement between DR-DEED Model and SGSD-DEED Model.

Emission reductions (Table 8): 66.90% (BC), 71.71% (C2), 69.57% (C3), and 62.68% (C4).

	Emissions			
	BC	C2	C3	C4
DR-DEED	$2.447\,404 \times 10^4$	$3.142\,655 \times 10^4$	$2.110\,633 \times 10^4$	$2.147\,348 \times 10^4$
SGSD-DEED	$8.093\,74 \times 10^3$	$8.901\,93 \times 10^3$	$6.425\,69 \times 10^3$	$8.000\,68 \times 10^3$
Percentage%	6.690×10^1	7.171	6.957×10^1	6.268×10^1

Table 8: Comparison of Emissions Improvement between DR-DEED Model and SGSD-DEED Model.

Total power generation is reduced by 53–55% across all weight cases (Table 9).

Scenario 2 shows consistent gains; the model progression analysis (Section 4.4) confirms that expanded customer flexibility drives the dominant improvement.

4.2 Experiment 2: Scalability Analysis

We assess scalability by varying customers over $[10, 15, 20]$ and generators over $[4, 6, 8]$. Each configuration was solved three times with averaged results, under BC weights.

Data. Customer and generator parameters were generated by fitting normal distributions to the original data from [1] and sampling new data points, with utility function parameters from [8].

Fig. 3 shows cost as a function of customer count for 4, 6, and 8 generators. Cost generally decreases as customer count increases, then stabilizes. Systems with fewer generators (4 or 6) benefit more from increased customer participation, while systems with more generators (8) show diminishing returns due to higher fixed operational costs.

Emissions (Fig. 4) follow a similar trend: decreasing with customer count and increasing with generator count. Fig. 5 shows utility as a function of customer count.

	Power Generated			
	BC	C2	C3	C4
DR-DEED	24266.59	2.426 659	$2.417\,639 \times 10^4$	$2.429\,11 \times 10^4$
SGSD-DEED	11259.04	1.094 040	$1.103\,150 \times 10^4$	$1.102\,217 \times 10^4$
Percentage%	53.63	5.492	5.433×10^1	5.459×10^1

Table 9: Comparison of Power Generated Improvement between DR-DEED Model and SGSD-DEED Model.

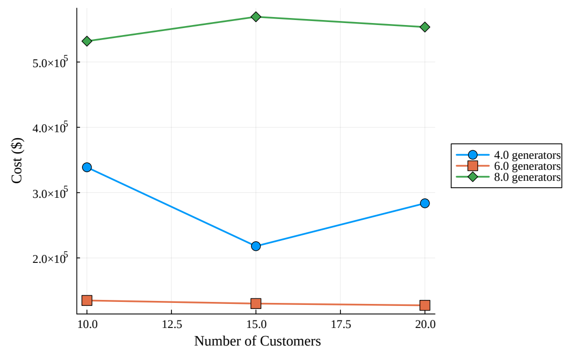


Figure 3: Cost, $\mathcal{C}$, for different numbers of generators.

Per-customer utility decreases as the customer base grows due to budget competition. The 8-generator configuration achieves the highest utility at smaller customer counts.

Losses (Fig. 6) decrease with customer count and increase with generator count. Load reduction (Fig. 7) increases with customer count. Across all metrics, systems with 4–6 generators benefit most from increased customer participation; the 4-generator, 15–20 customer configuration achieves the best overall performance. The CV across 3 trials ranges from 6% to 37%, with higher variance in emissions than cost due to sensitivity to stochastic customer initialization.

4.3 Experiment 3: IEEE 30-Bus Network Validation

We augment the SGSD-DEED formulation with DC power flow constraints (equations (31)–(34)) and solve it on the IEEE 30-bus benchmark network (30 buses, 41 branches). Both SC1 and SC2 are solved under BC weights, comparing the unconstrained solution ("No Network") against the DC-OPF–constrained solution ("IEEE 30-Bus DC-OPF").

Table 10 reports the results. In SC1, enforcing transmission constraints increases fuel cost by 4.9% (from $1.421\,58 \times 10^5$ to $1.490\,97 \times 10^5$) and emissions by 15.1% (from 8.009×10^3 to 9.220×10^3), while customer utility decreases by 27.8%. In SC2, the cost premium is 2.8% and the emission increase is 12.6%, with utility decreasing by 21.0%. These increases represent the *price of network feasibility*—the additional cost required to ensure that all branch flows remain within thermal limits.

Table 10: Impact of IEEE 30-bus DC-OPF constraints on SGSD-DEED (BC weights).

Scenario	**Configuration**	**Cost ($)**	**Emission (lb)**	**Utility ($)**
SC1	No Network	$1.421\,58 \times 10^5$	8.009×10^3	$1.592\,070 \times 10^6$
	IEEE 30-Bus DC-OPF	$1.490\,97 \times 10^5$	9.220×10^3	$1.149\,930 \times 10^6$
	Change (%)	+4.9%	+15.1%	−27.8%
SC2	No Network	$5.456\,53 \times 10^5$	5.6805×10^4	$5.566\,870 \times 10^6$
	IEEE 30-Bus DC-OPF	$5.607\,33 \times 10^5$	6.3953×10^4	$4.400\,590 \times 10^6$
	Change (%)	+2.8%	+12.6%	−21.0%

Fig. 8 shows the cost and emission comparison for SC1. The modest cost premium (3–5%) confirms economic viability under transmission constraints; the emission increase (13–15%) results from network-constrained dispatch of higher-emission generators.

Fig. 9 displays the branch utilization across the 41 transmission lines. Several branches operate near their thermal limits, indicating that the network constraints are binding and meaningfully restrict the dispatch

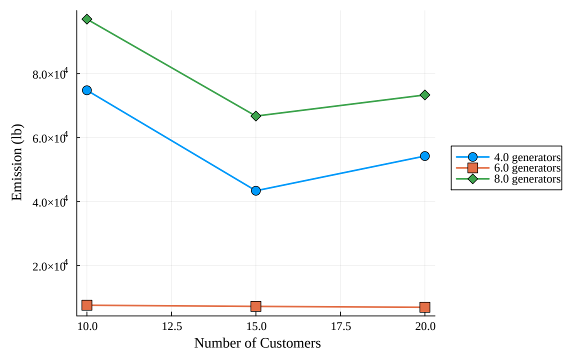


Figure 4: Emission, $\mathcal{E}$, for Different Generators.

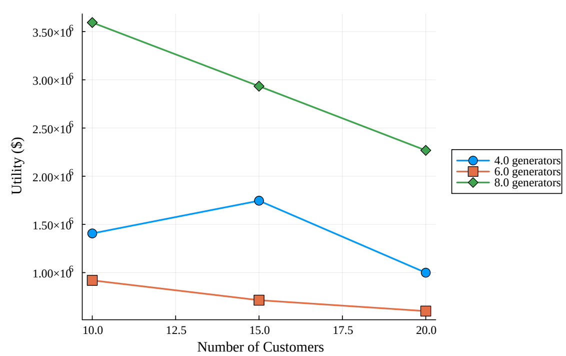


Figure 5: Utility, $\mathcal{U}$, for Different Generators.

solution. The solve time increases from 1.2 s to 4.6 s for SC1 and from 6.2 s to 10.3 s for SC2, confirming that the DC-OPF formulation remains computationally tractable. Across 3 trials, the cost difference between the two configurations is statistically significant at the 2.6σ level (SC1) and 4.7σ level (SC2), with the DC-OPF coefficient of variation approximately doubling relative to the unconstrained case (2.0% vs. 0.7% for SC1).

4.4 Experiment 4: Model Progression Analysis

Three model variants are solved on identical data under BC weights: (i) the baseline `DEED` (no demand response, $w_3 = 0$), (ii) the `DR-DEED` model of [1] (curtailment only), and (iii) the full `SGSD-DEED` (curtailment, shifting, storage, local generation, grid exchange). This progression demonstrates the incremental value of expanded customer flexibility, which is the core contribution of this paper.

Table 11 summarizes the results. In SC1, moving from `DEED` to `DR-DEED` yields a modest 7.9% cost reduction (from $3.173\,02 \times 10^5$ to $2.921\,51 \times 10^5$), reflecting the limited impact of curtailment alone. The full `SGSD-DEED` achieves a 55.8% cost reduction (to $1.402\,63 \times 10^5$) and a 71.6% emission reduction (from 2.7743×10^4 to 7.869×10^3). In SC2, the pattern is similar: `DR-DEED` provides only a 6.0% cost improvement over `DEED`, while `SGSD-DEED` achieves 47.7% cost reduction and 76.6% emission reduction. The bulk of the improvement comes from expanded customer flexibility—demand shifting, storage, distributed generation, and grid exchange.

Fig. 10 visualizes the cost and emission progression for SC1. The bar charts clearly show that the `DEED` $\rightarrow$ `DR-DEED` step provides incremental improvement, while the `DR-DEED` $\rightarrow$ `SGSD-DEED` step delivers the dominant gains. Expanded customer flexibility, not basic curtailment alone, drives the performance improvement. Across 3 trials, the `DEED` and `DR-DEED` solutions exhibit near-zero variance (CV $\approx$ 0%), reflecting the deterministic nature of the supply-side optimization, while `SGSD-DEED` shows modest variance (CV 0.9% for SC1, 4.1% for SC2) due to the richer solution space introduced by customer-side decision variables.

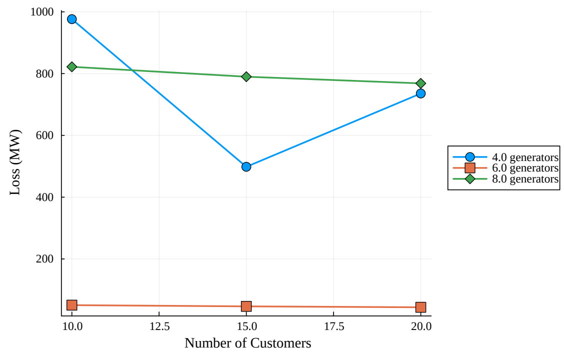


Figure 6: Power Loss.

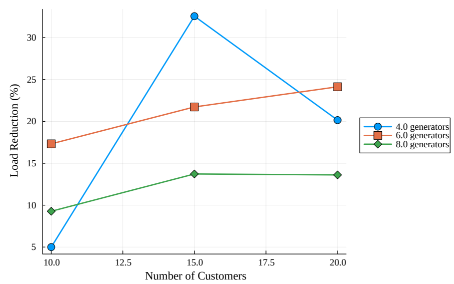


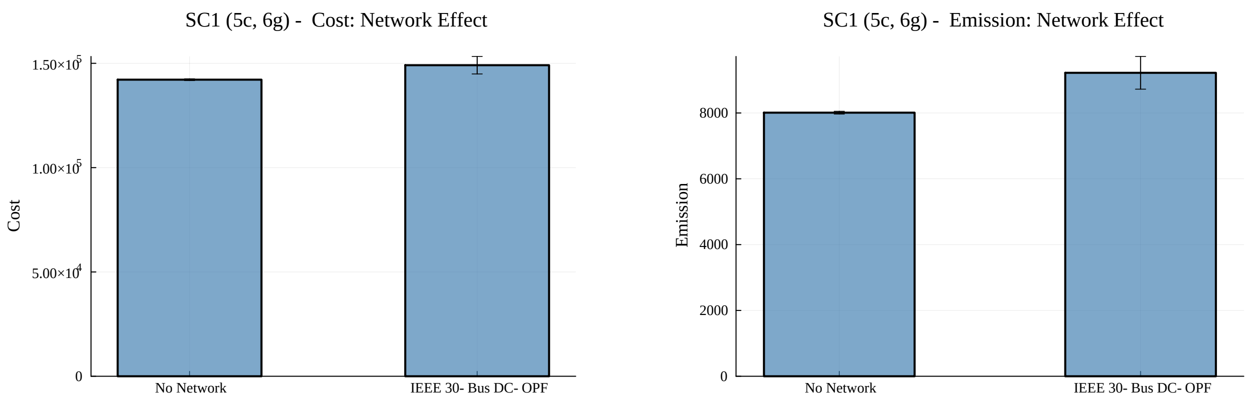
Figure 7: Power Load Reduction for Different Generators.

4.5 Experiment 5: Sensitivity and Robustness Analysis

Sensitivity analysis examines four dimensions: weight factors, customer willingness, customer count, and storage capacity.

4.5.1 Weight Factor Sensitivity

We evaluate 64 weight combinations (w_1, w_2, w_3) sampled uniformly on the unit simplex ($w_1 + w_2 + w_3 = 1$, $w_k \geq 0$) for SC1. Fig. 11 shows the resulting cost surface as a function of w_1 and w_2 (with $w_3 = 1-w_1-w_2$). The surface is smooth and monotonic: cost increases with w_1 and decreases with w_3, consistent with Propositions 6 and 2. The total cost range across all 64 weight combinations is only \$8,264 (CV = 1.5%), indicating that the model is robust to weight selection. Cost and emission are strongly negatively correlated ($r = -0.74$), confirming



(a) Fuel cost comparison. (b) Emission comparison.

Figure 8: Cost and emission under IEEE 30-bus DC-OPF constraints (SC1, BC weights).

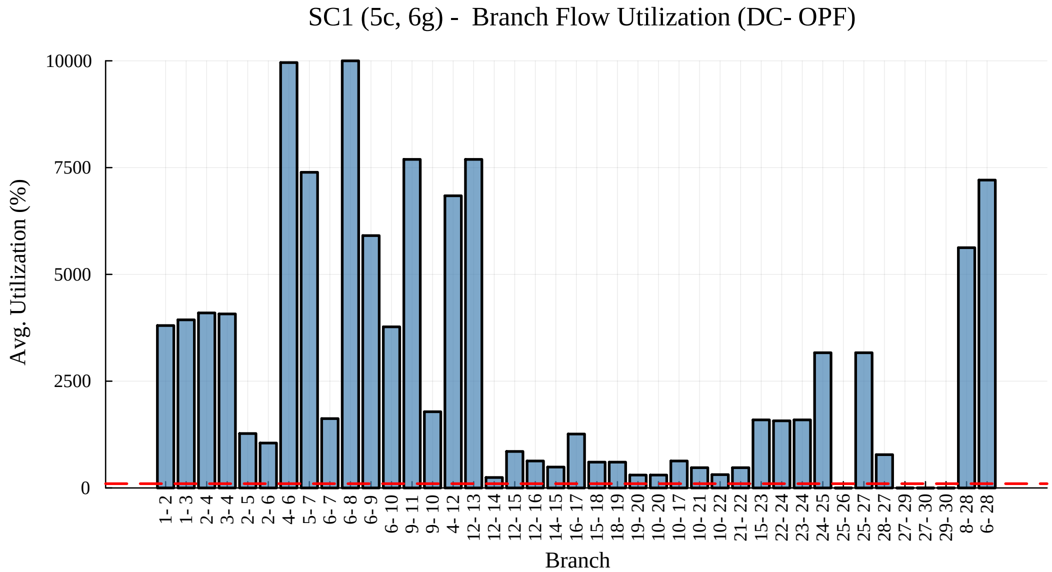
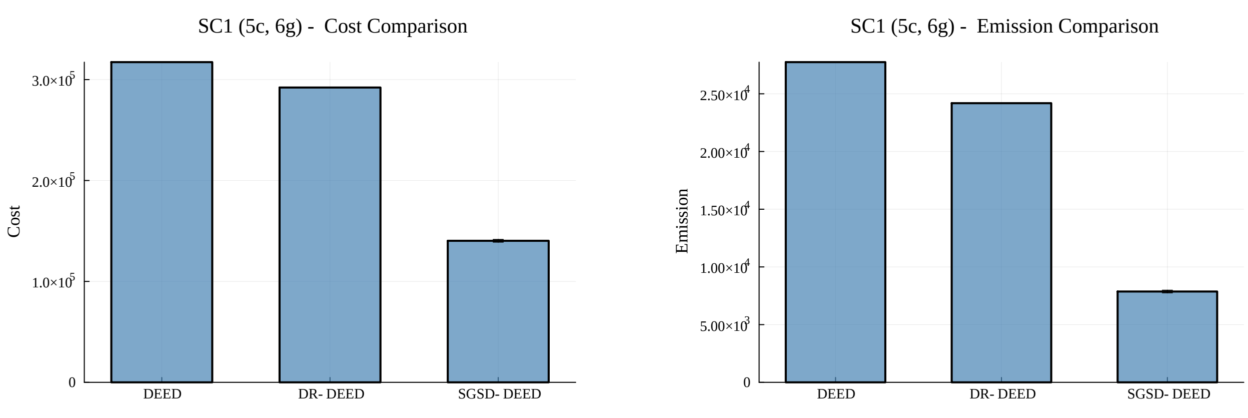


Figure 9: Branch utilization across 41 transmission lines under DC-OPF (SC1).

Table 11: Model progression: `DEED` $\rightarrow$ `DR-DEED` $\rightarrow$ `SGSD-DEED` (BC weights).

Scenario	Model	Cost ($)	Emission (lb)	Loss (MW)	Utility ($)
SC1	`DEED`	$3.173\,02 \times 10^5$	2.7743×10^4	3.045×10^2	—
	`DR-DEED`	$2.921\,51 \times 10^5$	2.4186×10^4	2.619×10^2	7.1942×10^4
	`SGSD-DEED`	$1.402\,63 \times 10^5$	7.869×10^3	5.23×10^1	$1.598\,990 \times 10^6$
	Reduction (%)	-55.8%	-71.6%	-82.8%	
SC2	`DEED`	$1.052\,800 \times 10^6$	$2.498\,41 \times 10^5$	1.3316×10^3	—
	`DR-DEED`	$9.894\,58 \times 10^5$	$1.951\,47 \times 10^5$	1.1405×10^3	$1.020\,69 \times 10^5$
	`SGSD-DEED`	$5.510\,40 \times 10^5$	5.8514×10^4	2.632×10^2	$4.595\,200 \times 10^6$
	Reduction (%)	-47.7%	-76.6%	-80.2%	



(a) Fuel cost progression.

(b) Emission progression.

Figure 10: Model progression: `DEED` $\rightarrow$ `DR-DEED` $\rightarrow$ `SGSD-DEED` (SC1, BC weights).

an inherent cost-emission tradeoff, while cost and utility are positively correlated ($r = +0.53$), reflecting the complementarity between economic efficiency and customer welfare in the Stackelberg framework.

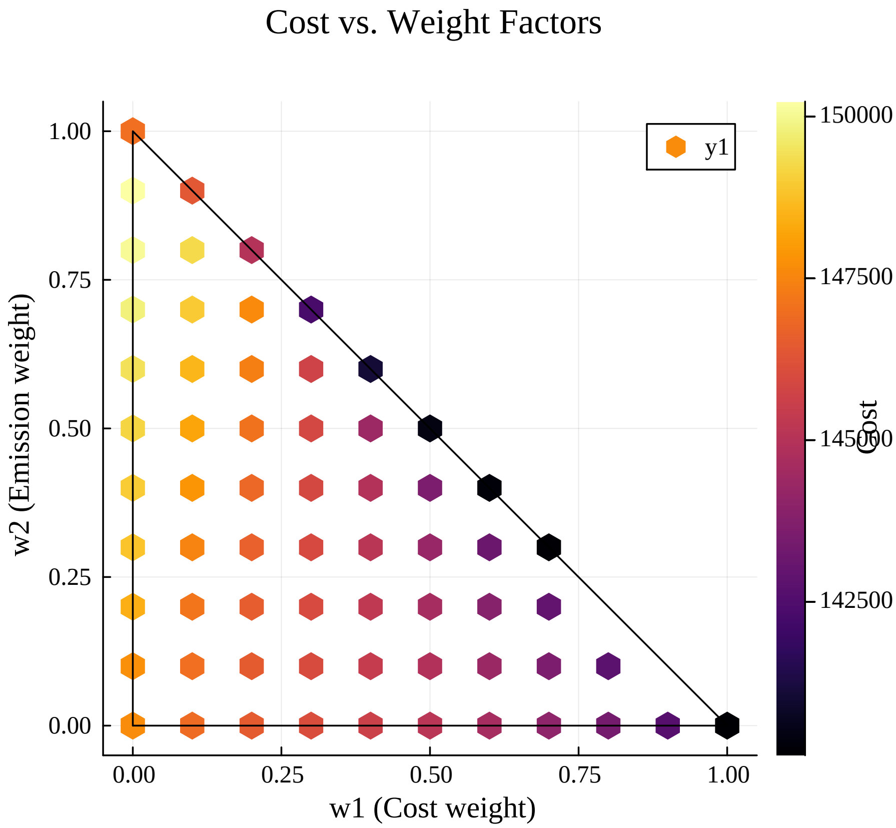


Figure 11: Fuel cost surface over 64 weight combinations on the unit simplex (SC1).

4.5.2 Customer Willingness (θ) Sensitivity

The willingness parameter $\theta_i \in [0, 1]$ modulates each customer's discomfort penalty for demand curtailment. We vary θ uniformly across all customers from 0.1 to 1.0 in increments of 0.1 under SC1, BC weights. Fig. 12 shows that both cost and emissions decrease monotonically as willingness increases, with the steepest improvement occurring in the range $\theta = 0.3$–0.7. Beyond $\theta = 0.8$, diminishing returns set in. The total cost reduction from $\theta = 0.1$ to $\theta = 1.0$ is 3.0% ($145,494 to $141,183), while customer utility increases by 13.3%—higher willingness primarily benefits customers.

4.5.3 Customer Count Scalability

We scale from 5 to 50 customers (steps of 5) using data drawn from the distributions of Experiment 2. Each configuration is solved 3 times and averaged. Table 12 and Fig. 13 present the results.

Cost decreases by 15.1% as customers grow from 5 to 50, with most gains realized by 30 customers—beyond which diminishing returns set in. Emissions follow a similar pattern, decreasing by 23.3%. Solve time scales from 1.4 s (5 customers) to approximately 12 s (50 customers), confirming computational tractability at moderate scale. The per-customer cost decreases from $28,470 ($n = 5$) to $2,417 ($n = 50$), exhibiting strong economies of scale in demand response participation. Non-monotonicities at $n = 30$ and $n = 40$ (1.7% and 2.4%) fall within inter-trial variance (0.27σ and 0.55σ). The per-customer utility decreases as the customer base grows, reflecting competition for limited incentive budgets.

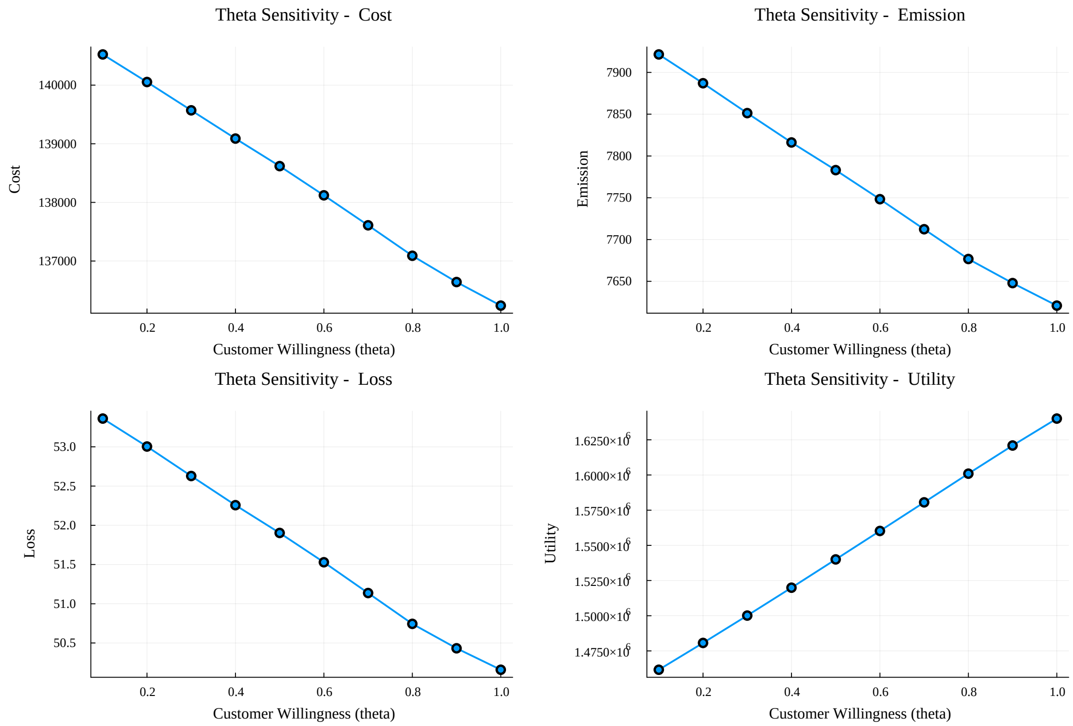


Figure 12: Sensitivity of cost, emission, loss, and utility to customer willingness θ (SC1, BC weights).

Table 12: Scalability analysis: 5 to 50 customers (SC1 generators, BC weights).

Customers	**Cost (\$)**	**Emission (lb)**	**Loss (MW)**	**Time (s)**
5	1.42349×10^5	8.046×10^3	5.43×10^1	1.4
1.0×10^1	1.34842×10^5	7.621×10^3	5.04×10^1	1.8
1.5×10^1	1.30126×10^5	7.266×10^3	4.65×10^1	2.1
2.0×10^1	1.24457×10^5	6.641×10^3	3.98×10^1	2.5
2.5×10^1	1.23641×10^5	6.547×10^3	3.89×10^1	3.1
3.0×10^1	1.25701×10^5	6.508×10^3	3.94×10^1	4.3
3.5×10^1	1.21035×10^5	6.197×10^3	3.54×10^1	5.5
4.0×10^1	1.23898×10^5	6.252×10^3	3.69×10^1	7.6
4.5×10^1	1.20874×10^5	6.176×10^3	3.51×10^1	1.28×10^1
5.0×10^1	1.20833×10^5	6.174×10^3	3.51×10^1	1.20×10^1

4.5.4 Storage Capacity and Parameter Importance

We scale storage capacity from 0.25× to 5× baseline. A phase transition occurs at 0.5–0.75×: below this threshold, storage is insufficient for temporal arbitrage and the solver converges to a supply-side-only corner solution (cost \$30,480, utility \$5,086, zero losses). Above 0.75×, cost stabilizes near \$140,000–\$144,000 with diminishing returns beyond 1.5×: each doubling reduces cost by less than 1%, while utility continues to grow (+16.6% from 3× to 5×). To identify the most influential parameters, Fig. 14 presents a tornado diagram showing the sensitivity of total cost to one-at-a-time perturbations of the key model parameters. Storage capacity exhibits the widest total range, driven largely by the degenerate corner solution below 0.5× baseline (see above); excluding that regime, the weight factor w_1 and customer count show the greatest practical sensitivity, followed by customer willingness θ. This ranking provides practitioners with guidance on where to focus modeling effort and data collection.

4.5.5 Pareto Front Analysis

To characterize the cost–emission trade-off more precisely, we apply the ε-constraint method [43]: minimize cost subject to an emission upper bound $E \leq \varepsilon$, sweeping ε uniformly from the minimum achievable emission ($E_{\min} = 6{,}409$ lb, obtained by minimizing emission alone) to the emission at minimum cost ($E_{\max} = 8{,}964$ lb).

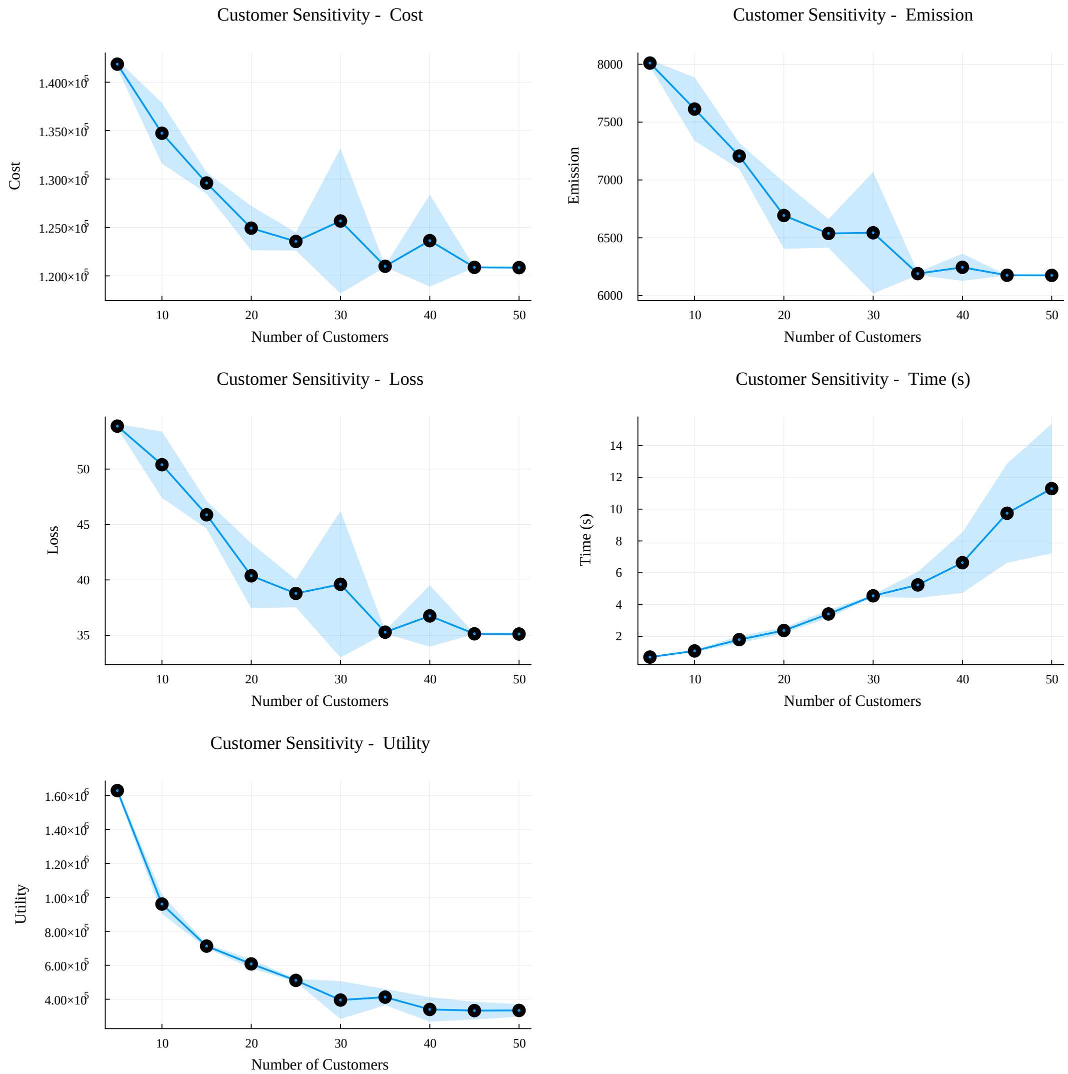


Figure 13: Cost, emission, loss, and utility as a function of customer count (SC1, BC weights). Shaded bands indicate ±1 standard deviation across 3 trials.

This approach generates uniformly distributed Pareto points regardless of frontier convexity, complementing the weighted-sum results of Experiment 5A.

Fig. 15 displays the resulting Pareto front for SC1 (30 points); we focus on SC1 as a representative case for this methodological demonstration, as the trade-off structure is qualitatively similar across scenarios. The frontier is smooth and convex, consistent with the feasible set convexity established in Proposition 2. The trade-off is modest: reducing emissions from 8,964 to 6,409 lb (a 28% reduction) increases cost from \$137,660 to \$143,853 (a 4.5% increase). Customer utility remains stable at approximately \$1.31M across the entire frontier, indicating that the cost–emission trade-off does not substantially compromise customer welfare.

Comparing the ε-constraint and weighted-sum approaches (Fig. 16), both methods trace the same frontier, but their sampling differs: weighted-sum points cluster near the extremes (particularly at low cost/high emission), while ε-constraint provides uniform coverage. This confirms that the weighted-sum scalarization used throughout this paper correctly identifies the Pareto-optimal region, while the ε-constraint analysis provides a more complete characterization of the trade-off surface. The marginal rate of substitution $|dC/dE|$ ranges from approximately \$2.1/lb at the high-emission end to \$4.8/lb at the low-emission end, indicating that the first units of emission reduction are relatively inexpensive, with diminishing returns as the frontier approaches $E_{\min}$.

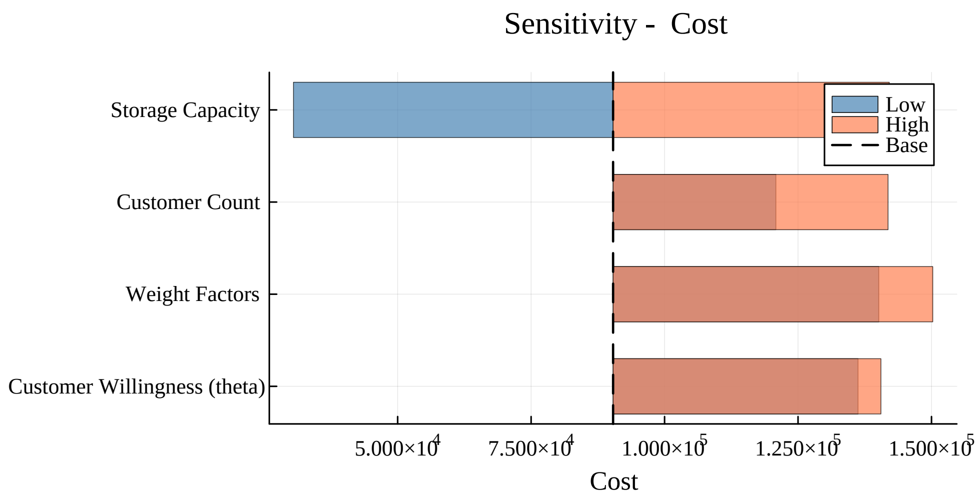


Figure 14: Tornado diagram: parameter importance for total fuel cost (SC1, BC weights). Each bar spans the cost range observed when one parameter is swept over its full experimental range while others are held at baseline; the dashed line marks the midrange reference value.

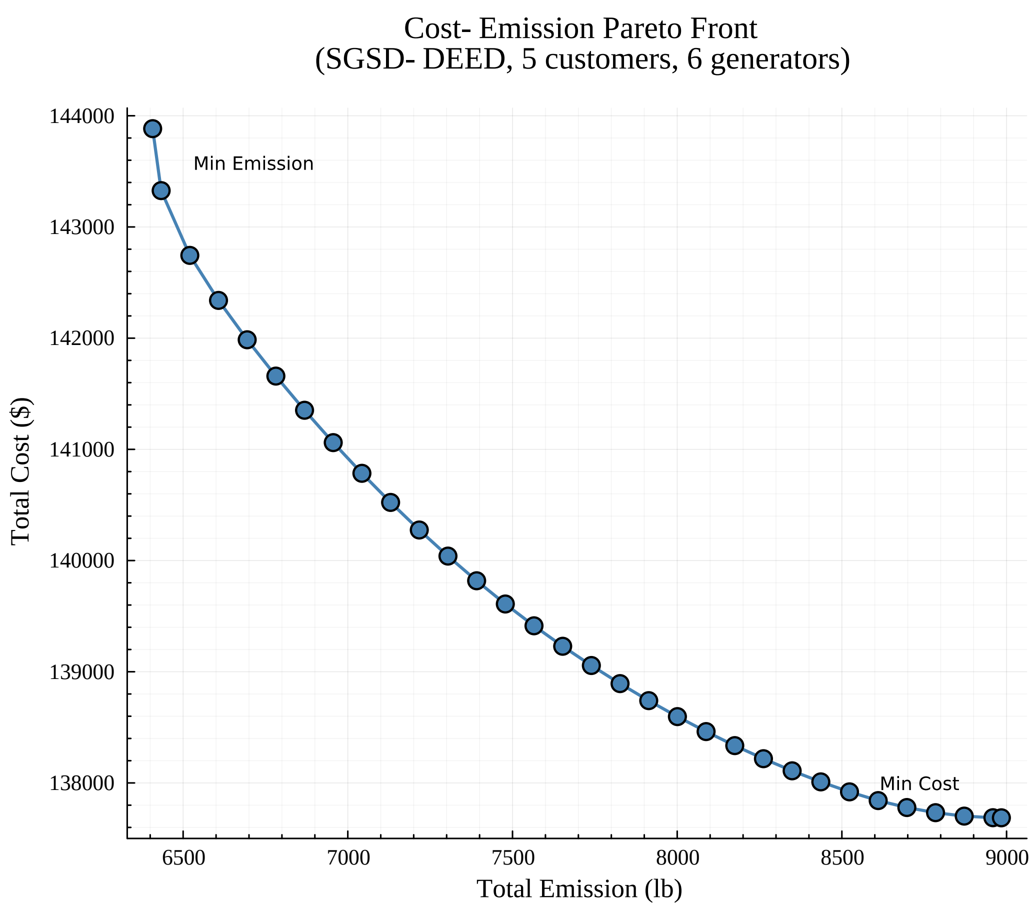


Figure 15: Cost–emission Pareto front via ε-constraint method (SC1, 30 points).

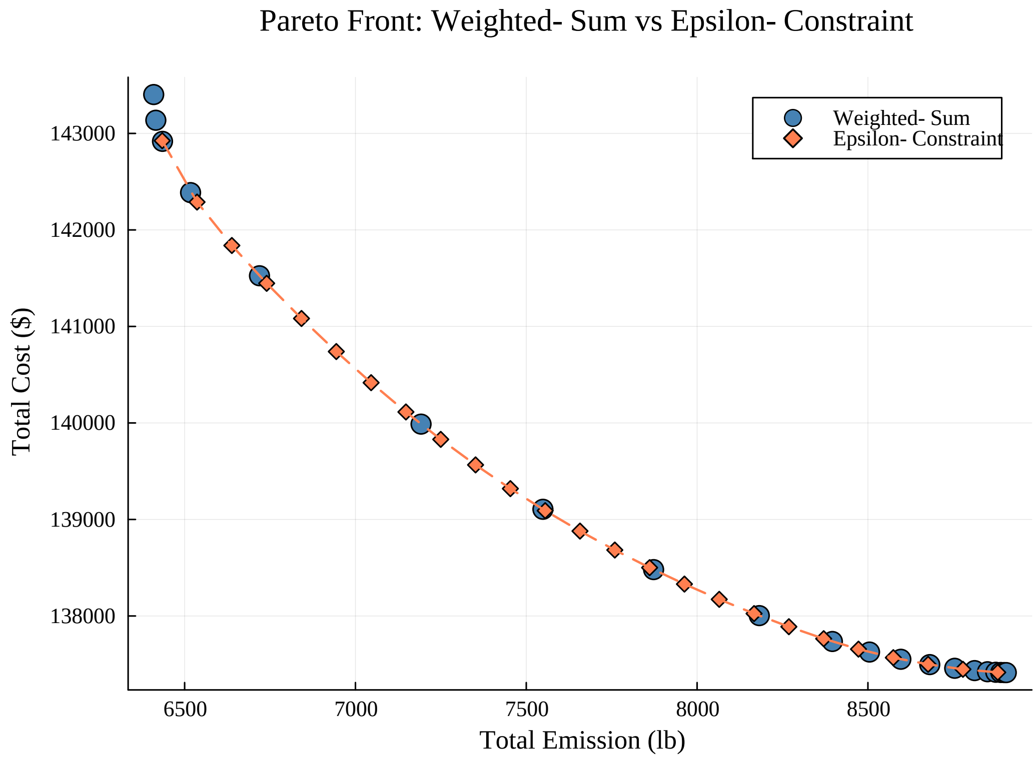


Figure 16: Pareto front comparison: weighted-sum (20 points) vs. ε-constraint (25 points).

4.6 Experiment 6: Saudi Eastern Province Case Study

We construct a case study using parameters representative of the Saudi Eastern Province summer grid.

4.6.1 Saudi Grid Data

The Saudi grid parameters differ from the PJM-based data in three respects:

(i) A single sharp afternoon peak driven by air-conditioning loads, which constitute approximately 70% of Saudi residential electricity consumption [44]. We construct a representative 24-hour profile calibrated to published aggregate statistics from the SEC Annual Report [45] and the Saudi Open Data Portal [46] (national peak load: 70,400 MW; Eastern Province annual consumption: 86,903 GWh): demand ranges from 590 MW (hour 4) to 1,300 MW (hour 14), yielding a peak-to-minimum ratio of 2.20 and a load factor of ~ 0.58.

(ii) A gas-dominated generator fleet comprising 3 combined-cycle gas turbines (CCGT), 2 simple-cycle gas turbines (SCGT), and 1 oil-fired steam unit for the 6-unit system, with cost and emission coefficients adapted from [47] and EPA AP-42 emission factors [48] to reflect the Saudi fuel mix. The gas-heavy fleet yields lower per-MWh emissions than the mixed-fuel fleet of the PJM-based scenarios.

(iii) A constructed time-of-use (TOU) tariff with off-peak rate of 48 \$/MWh and peak rate of 195–203 \$/MWh during hours 14–17, producing a 4:1 peak-to-off-peak ratio—substantially higher than the $\sim$ 2–3:1 ratio in typical PJM wholesale markets. The current SEC regulated tariff is flat/tiered [45]; we adopt this illustrative TOU structure to assess the model's response to strong peak pricing signals, consistent with demand response program designs proposed in the literature [22].

Customer discomfort coefficients (K_1) are assumed approximately 15% higher than in the PJM-based scenarios, reflecting the greater discomfort associated with air-conditioning curtailment in extreme heat ($\sim 45°$C ambient temperature); this is a physical input assumption rather than an algorithmic recalibration. All remaining customer parameters—willingness thresholds (θ), storage capacities, budget limits, and production coefficients—are held at the same values as in the PJM-based scenarios. The network topology (IEEE 30-bus), weight schemes, and solution methodology remain identical to those of Experiments 3 and 4, isolating the effect of the regional data parameters.

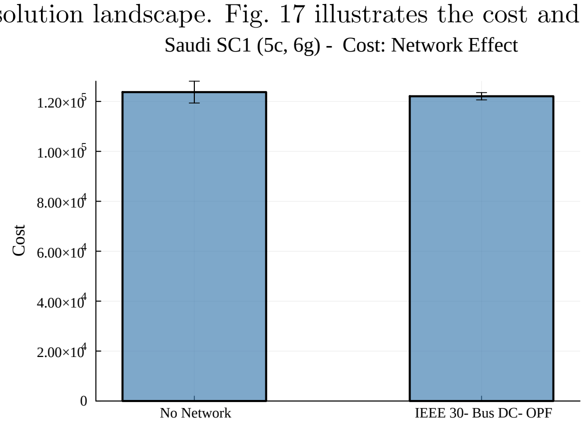
4.6.2 Network Effect with Saudi Data

Following the same methodology as Experiment 3, we solve SGSD-DEED with and without IEEE 30-bus DC-OPF constraints under BC weights for both SC1 ($n = 5$, $J = 6$) and SC2 ($n = 7$, $J = 10$), using the Saudi data parameters described above with 3 trials per configuration. Table 13 reports the results.

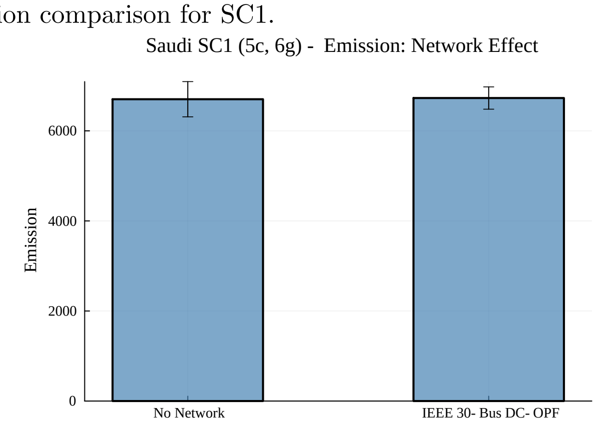
Table 13: Impact of IEEE 30-bus DC-OPF constraints on SGSD-DEED with Saudi data (BC weights).

Scenario	Configuration	Cost ($)	Emission (lb)	Loss (MW)	Utility ($)
SC1	No Network	1.23733×10^5	6.703×10^3	4.34×10^1	8.48744×10^5
	IEEE 30-Bus DC-OPF	1.22081×10^5	6.729×10^3	1.116×10^2	8.28990×10^5
	Change (%)	−1.3%	+0.4%	+157%	−2.3%
SC2	No Network	2.13014×10^5	1.1527×10^4	2.446×10^2	1.254430×10^6
	IEEE 30-Bus DC-OPF	2.17108×10^5	1.2231×10^4	1.644×10^2	1.096890×10^6
	Change (%)	+1.9%	+6.1%	−33%	−12.6%

In contrast to the PJM-based scenarios (section 4.3), where DC-OPF constraints increased cost by 3–5%, the Saudi data shows mixed network effects: a slight cost *decrease* in SC1 (−1.3%) but a modest increase in SC2 (+1.9%). Emissions increase modestly (+0.4% to +6.1%), indicating a cost-emission tradeoff under the gas/oil fuel mix that differs qualitatively from PJM. Network constraint effects are region-dependent. The solver variance under Saudi data is lower than under PJM data (CV $\leq$ 1.2% vs. up to 2.0%), suggesting that the sharper peak structure of the Saudi demand profile yields a more constrained and therefore more reproducible solution landscape. Fig. 17 illustrates the cost and emission comparison for SC1.



(a) Fuel cost comparison.



(b) Emission comparison.

Figure 17: Cost and emission under IEEE 30-bus DC-OPF constraints with Saudi data (SC1, BC weights).

Fig. 18 shows the branch utilization across the 41 transmission lines. The Saudi demand profile produces a congestion pattern distinct from the PJM-based case (Fig. 9): the sharper afternoon peak concentrates stress on fewer branches during hours 14–17, while off-peak branches are less utilized. This difference further illustrates the region-dependent nature of network constraint effects.

4.6.3 Model Progression with Saudi Data

Following the same methodology as Experiment 4, we solve DEED, DR-DEED, and SGSD-DEED on shared Saudi data under BC weights, with 3 trials per model. Table 14 reports the results.

The progression is monotonic: DEED > DR-DEED > SGSD-DEED for cost, emission, and loss in both scenarios. The SGSD-DEED achieves 46–47% cost reduction versus baseline DEED, and emission reductions of 68–70%. These results are consistent with the PJM-based findings of section 4.4 (48–56% cost, 72–77% emission reduction), indicating that the model's performance generalizes across grid contexts. The somewhat lower emission reduction in

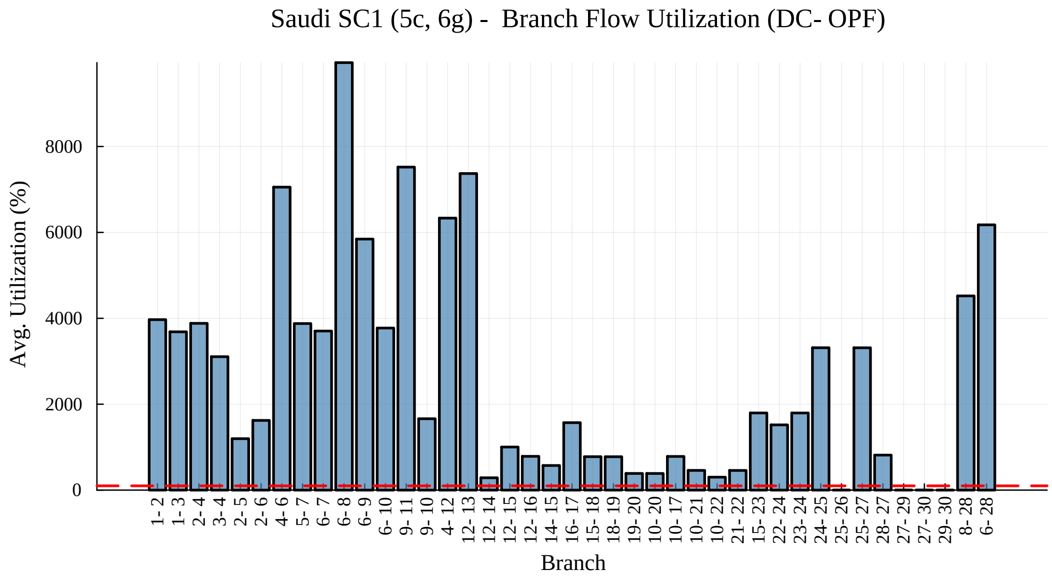
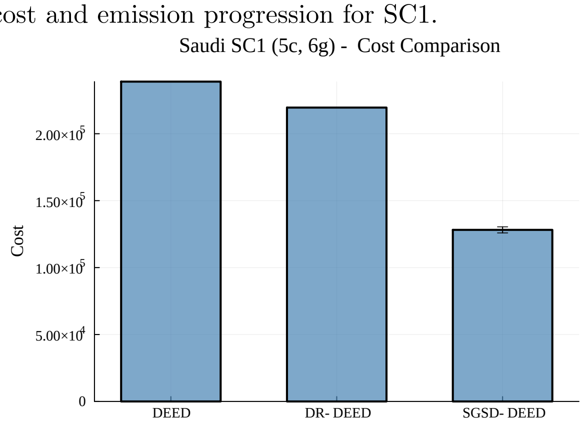


Figure 18: Branch utilization under DC-OPF with Saudi data (SC1).

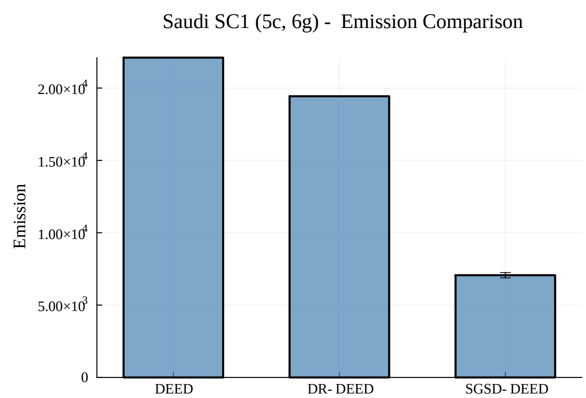
Table 14: Model progression with Saudi data: `DEED` $\rightarrow$ `DR-DEED` $\rightarrow$ `SGSD-DEED` (BC weights).

Scenario	Model	Cost ($)	Emission (lb)	Loss (MW)	Utility ($)
SC1	`DEED`	$2.389\,07 \times 10^5$	2.2103×10^4	2.226×10^2	—
	`DR-DEED`	$2.195\,23 \times 10^5$	1.9434×10^4	1.881×10^2	$1.506\,90 \times 10^5$
	`SGSD-DEED`	$1.281\,76 \times 10^5$	7.066×10^3	4.78×10^1	$6.620\,00 \times 10^5$
	Reduction (%)	−46.4%	−68.0%	−78.5%	
SC2	`DEED`	$4.180\,23 \times 10^5$	3.9642×10^4	1.2550×10^3	—
	`DR-DEED`	$3.862\,37 \times 10^5$	3.5172×10^4	1.0699×10^3	$2.196\,46 \times 10^5$
	`SGSD-DEED`	$2.220\,67 \times 10^5$	1.1807×10^4	2.638×10^2	$7.853\,39 \times 10^5$
	Reduction (%)	−46.9%	−70.2%	−79.0%	

relative terms reflects the gas-dominated Saudi fleet's lower baseline emission intensity. Fig. 19 visualizes the cost and emission progression for SC1.



(a) Fuel cost progression.



(b) Emission progression.

Figure 19: Model progression with Saudi data: `DEED` $\rightarrow$ `DR-DEED` $\rightarrow$ `SGSD-DEED` (SC1, BC weights).

Notably, `DR-DEED` provides only 8.1% (SC1) and 7.6% (SC2) cost reduction over `DEED`, reinforcing the finding from section 4.4 that basic curtailment alone is insufficient regardless of grid context. The SEC tariff's 4:1 peak ratio creates a substantially stronger demand response signal during hours 14–17 than typical wholesale markets, yet the model adapts without algorithmic recalibration (the only modified input, K_1, reflects physical conditions rather than solver tuning). As in Experiment 4, the `DEED` and `DR-DEED` solutions are deterministic (CV = 0%), while SGSD-DEED exhibits modest solver variance (CV = 2.5% for SC1 and 2.9% for SC2), comparable to the PJM-based results. The SGSD-DEED framework generalizes across grid contexts. A full Saudi sensitivity analysis is deferred to future work.

4.7 Remark: Comparison with Metaheuristic Methods

We benchmarked NSGA-II (population 100, 200 generations) against Ipopt on both scenarios. NSGA-II found 1–2 Pareto points vs. Ipopt's 20, with 48× lower hypervolume in SC1. In SC2 (3,110 decision variables), NSGA-II failed to produce any feasible solutions due to constraint violations. This confirms that the high-dimensional (1,700–3,100 variables), heavily-constrained structure of SGSD-DEED favors gradient-based interior-point methods over metaheuristic search, justifying our solver methodology.

5 Conclusion

The SGSD-DEED model is an integrated game-theoretic and multi-objective optimization approach for the Dynamic Economic Emission Dispatch (DEED) problem in smart grids with demand response (DR). The model was formulated as a multi-period Stackelberg game with proven equilibrium existence (Proposition 1), convex feasible region (Proposition 2), and weak Pareto optimality guarantees for the weighted-sum scalarization (Proposition 6). The DC structure of the objective (Proposition 3) was characterized, and the computational complexity was analyzed (Remark 4).

Compared to the DR-DEED model of [1], the SGSD-DEED model achieves up to 80% reduction in system losses, 53% reduction in fuel costs, and 72% reduction in emissions on the original test scenarios (Experiment 1). A model progression analysis (Experiment 4) demonstrated that expanded customer flexibility—rather than basic curtailment alone—drives the dominant gains, achieving up to 56% cost reduction and 77% emission reduction relative to the baseline DEED model. Validation on the IEEE 30-bus benchmark with DC power flow constraints (Experiment 3) confirmed that the model remains viable under realistic transmission limits, with a modest 3–5% cost premium representing the price of network feasibility.

Scalability analysis showed that the model adapts effectively to varying numbers of customers and generators, with solve times remaining under 13 s for 50 customers (Experiment 5D) and per-customer cost exhibiting strong economies of scale ($28,470 at $n = 5$ to $2,417 at $n = 50$). Sensitivity analysis across 64 weight combinations, 10 willingness levels, and 8 storage capacity scales confirmed robust, well-behaved trade-offs: total cost varies by only CV = 1.5% across all weight combinations, cost and emission exhibit a strong tradeoff ($r = -0.74$), and a storage capacity phase transition below 0.5× baseline was identified, below which customer flexibility mechanisms are inactive (Experiment 5). The tornado analysis identified weight factors and customer willingness as the most influential parameters, providing practitioners with guidance on where to focus modeling and data collection efforts. Pareto front analysis via the ε-constraint method confirmed that the cost–emission trade-off is modest: a 28% emission reduction requires only a 4.5% cost increase, and the frontier is uniformly sampled by the ε-constraint approach while weighted-sum points cluster near the extremes.

A Saudi Eastern Province case study (Experiment 6) validated the model under representative Middle Eastern grid conditions characterized by a sharp afternoon air-conditioning peak, a gas-dominated generator fleet, and a constructed time-of-use tariff with a 4:1 peak-to-off-peak ratio. The SGSD-DEED achieved 46–47% cost reduction and 68–70% emission reduction versus baseline DEED, consistent with the PJM-based results. A particularly notable finding is that network constraint effects are qualitatively region-dependent: DC-OPF constraints increase cost by 3–5% under PJM conditions but show mixed effects under Saudi conditions (−1.3% in SC1, +1.9% in SC2), while shifting the cost-emission tradeoff in opposite directions. This underscores that transmission-aware demand response program design cannot rely on universal assumptions about network constraint impacts, and confirms that the SGSD-DEED framework generalizes across grid contexts without algorithmic recalibration.

Future work includes real-time implementation with stochastic renewables, deep learning-based demand forecasting, and AC power flow constraints with voltage stability.

Declaration of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data and code used to produce the results of this paper are publicly available. A Julia package, DRDeed.jl [41], implements the optimization models.

Full experiment scripts and result files are available at https://github.com/mmogib/NMS_SGSD_DEED_Paper.git.

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AI Use Declaration

During the preparation of this work the authors used Claude (Anthropic) in order to assist with manuscript editing, including tightening prose, verifying LaTeX formatting, and checking internal consistency of cross-references and notation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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References

[1] N. I. Nwulu and X. Xia, "Multi-objective dynamic economic emission dispatch of electric power generation integrated with game theory based demand response programs," *Energy Conversion and Management*, vol. 89, pp. 963–974, 2015. DOI: `https://doi.org/10.1016/j.enconman.2014.11.001`.

[2] S. Wang, S. Bi, and Y.-J. A. Zhang, "Demand response management for profit maximizing energy loads in real-time electricity market," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6387–6396, 2018. DOI: `10.1109/TPWRS.2018.2827401`.

[3] C. Eid, E. Koliou, M. Valles, J. Reneses, and R. Hakvoort, "Time-based pricing and electricity demand response: Existing barriers and next steps," *Utilities Policy*, vol. 40, pp. 15–25, 2016. DOI: `https://doi.org/10.1016/j.jup.2016.04.001`.

[4] T. Cui, Y. Wang, S. Yue, S. Nazarian, and M. Pedram, "A game-theoretic price determination algorithm for utility companies serving a community in smart grid," in *2013 IEEE PES Innovative Smart Grid Technologies Conference (ISGT)*, IEEE, 2013, pp. 1–6.

[5] I. Atzeni, L. G. Ordóñez, G. Scutari, D. P. Palomar, and J. R. Fonollosa, "Demand-side management via distributed energy generation and storage optimization," *IEEE Transactions on Smart Grid*, vol. 4, no. 2, pp. 866–876, 2012.

[6] I. Atzeni, L. G. Ordóñez, G. Scutari, D. P. Palomar, and J. R. Fonollosa, "Noncooperative and cooperative optimization of distributed energy generation and storage in the demand-side of the smart grid," *IEEE transactions on signal processing*, vol. 61, no. 10, pp. 2454–2472, 2013.

[7] S. Belhaiza and U. Baroudi, "A game theoretic model for smart grids demand management," *IEEE Transactions on Smart Grid*, vol. 6, no. 3, pp. 1386–1393, 2015. DOI: 10.1109/TSG.2014.2376632.

[8] S. Belhaiza, U. Baroudi, and I. Elhallaoui, "A game theoretic model for the multiperiodic smart grid demand response problem," *IEEE Systems Journal*, vol. 14, no. 1, pp. 1147–1158, 2019.

[9] F. Cui, A. Dou, and G. Zhang, "A game strategy for demand response based on load monitoring in smart grid," *Frontiers in Energy Research*, vol. 11, p. 1 240 542, 2023.

[10] M. Fochesato, C. Cenedese, and J. Lygeros, "A Stackelberg game for incentive-based demand response in energy markets," in *Proceedings of the 61st IEEE Conference on Decision and Control (CDC)*, IEEE, 2022. DOI: 10.48550/arXiv.2204.08730.

[11] J. Li, T. Li, and D. Dong, "Demand response management of smart grid based on Stackelberg-evolutionary joint game," *Science China Information Sciences*, vol. 66, no. 8, p. 182 201, 2023. DOI: 10.1007/s11432-022-3674-6.

[12] Y. Wan, J. Qin, Y. Shi, W. Fu, and F. Xiao, "Stackelberg–Nash game approach for price-based demand response in retail electricity trading," *International Journal of Electrical Power & Energy Systems*, vol. 155, p. 109 577, 2024. DOI: 10.1016/j.ijepes.2023.109577.

[13] Y. Wu *et al.*, "Energy management for PV prosumers inside microgrids based on Stackelberg–Nash game considering demand response," *Sustainable Energy Technologies and Assessments*, vol. 69, p. 103 882, 2024. DOI: 10.1016/j.seta.2024.103856.

[14] S. Xue, Y. Wang, Y. Liu, Z. Wang, Y. Wang, and D. Yang, "Bi-level optimization and Stackelberg game in multi-integrated energy systems," *Journal of Energy Storage*, vol. 111, p. 115 356, 2025. DOI: 10.1016/j.est.2025.115356.

[15] S. A. Ali *et al.*, "Optimizing smart grid demand response: A Stackelberg game framework for priority-aware dynamic pricing and load scheduling," *Engineering Reports*, vol. 7, no. 8, e70341, 2025. DOI: 10.1002/eng2.70341.

[16] K. Ullah *et al.*, "Demand side management strategy for multi-objective day-ahead scheduling considering wind energy in smart grid," *Energies*, vol. 15, p. 6900, Sep. 2022. DOI: 10.3390/en15196900.

[17] N. Nalini, C. Kumar, P. Vijayarajan, and N. Chidambararaj, "Integrated approach for dynamic economic/emission dispatch problem: Multi-objective moth flame optimizer with transmission loss prediction using cascaded forward neural network," *Electrical Engineering*, vol. 106, pp. 3495–3528, 2024.

[18] J. Soni and K. Bhattacharjee, "Multi-objective dynamic economic emission dispatch integration with renewable energy sources and plug-in electrical vehicle using equilibrium optimizer," *Environment, Development and Sustainability*, vol. 26, pp. 8555–8586, 2024.

[19] H. Y. Shahri, S. A. Hosseini, and J. Pourhossein, "Multi-objective evolutionary method for multi-area dynamic emission/economic dispatch considering energy storage and renewable energy units," *Heliyon*, vol. 10, no. 18, e37476, 2024. DOI: 10.1016/j.heliyon.2024.e37476.

[20] S. Guo, "Economic dispatch model of renewable energy system considering demand response," *International Journal of Renewable Energy Development*, vol. 14, no. 2, pp. 311–321, 2025. DOI: 10.61435/ijred.2025.60680.

[21] H. Kafshian and M. A. Saniee Monfared, "A multi-layer–multi-player game model in electricity market," *IET Generation, Transmission & Distribution*, vol. 18, no. 7, pp. 1494–1515, 2024. DOI: 10.1049/gtd2.13125.

[22] J. Ma and B. Venkatesh, "New real-time demand response market co-optimized with conventional energy market," *IEEE Systems Journal*, vol. 16, no. 4, pp. 6381–6392, 2022. DOI: 10.1109/JSYST.2021.3132786.

[23] X. Chen, "Enhance low-carbon power system operation via carbon-aware demand response," *IET Energy Internet*, e12004, 2024. DOI: 10.1049/ein2.12004.

[24] Z. Wang, M. V. Liu, G. C. Guo, and C. L. Anderson, "AC optimal power flow in power systems with renewable energy integration: A review of formulations and case studies," *IEEE Access*, vol. 11, pp. 103 744–103 766, 2023. DOI: 10.1109/ACCESS.2023.3314330.

[25] E. Afzalan and M. Joorabian, "Emission, reserve and economic load dispatch problem with non-smooth and non-convex cost functions using epsilon-multi-objective genetic algorithm variable," *International Journal of Electrical Power & Energy Systems*, vol. 52, pp. 55–67, 2013. DOI: https://doi.org/10.1016/j.ijepes.2013.03.017.

[26] A. Elaiw, X. Xia, and A. Shehata, "Hybrid de-sqp and hybrid pso-sqp methods for solving dynamic economic emission dispatch problem with valve-point effects," *Electric Power Systems Research*, vol. 103, pp. 192–200, 2013. DOI: https://doi.org/10.1016/j.epsr.2013.05.015.

[27] D. Aydin, S. Özyön, C. Yaşar, and T. Liao, "Artificial bee colony algorithm with dynamic population size to combined economic and emission dispatch problem," *International Journal of Electrical Power & Energy Systems*, vol. 54, pp. 144–153, 2014. DOI: https://doi.org/10.1016/j.ijepes.2013.06.020.

[28] S. Sayah, A. Hamouda, and A. Bekrar, "Efficient hybrid optimization approach for emission constrained economic dispatch with nonsmooth cost curves," *International Journal of Electrical Power & Energy Systems*, vol. 56, pp. 127–139, 2014. DOI: https://doi.org/10.1016/j.ijepes.2013.11.001.

[29] A. Arif and F. Javed, "Integrating renewables economic dispatch with demand side management in micro-grids: A genetic algorithm-based approach," *Energy Efficiency*, vol. 7, Apr. 2014. DOI: 10.1007/s12053-013-9223-9.

[30] C. L. Chinnadurrai and T. A. A. Victoire, "Enhanced multi-objective crisscross optimization for dynamic economic emission dispatch considering demand response and wind power uncertainty," *Soft Computing*, vol. 24, no. 12, pp. 9021–9038, 2020.

[31] A. Elaiw, X. Xia, and A. Shehata, "Application of model predictive control to optimal dynamic dispatch of generation with emission limitations," *Electric power systems research*, vol. 84, no. 1, pp. 31–44, 2012.

[32] M. Fahrioglu and F. L. Alvarado, "Designing incentive compatible contracts for effective demand management," *IEEE Transactions on power Systems*, vol. 15, no. 4, pp. 1255–1260, 2000.

[33] A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*, 3rd. John Wiley & Sons, 2014.

[34] J.-J. Laffont and D. Martimort, *The Theory of Incentives: The Principal-Agent Model*. Princeton University Press, 2002.

[35] A. Mas-Colell, M. D. Whinston, and J. R. Green, *Microeconomic Theory*. Oxford University Press, 1995.

[36] J. Nocedal and S. J. Wright, *Numerical Optimization* (Springer Series in Operations Research), 2nd. Springer, 2006.

[37] R. G. Jeroslow, "The polynomial hierarchy and a simple model for competitive analysis," *Mathematical Programming*, vol. 32, no. 2, pp. 146–164, 1985. DOI: 10.1007/BF01586088.

[38] A. Wächter and L. T. Biegler, *On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming*, Mar. 15, 2004.

[39] M. Lubin, O. Dowson, J. Dias Garcia, J. Huchette, B. Legat, and J. P. Vielma, "JuMP 1.0: Recent improvements to a modeling language for mathematical optimization," *Mathematical Programming Computation*, 2023. DOI: 10.1007/s12532-023-00239-3.

[40] J. Bezanson, A. Edelman, S. Karpinski, and V. B. Shah, "Julia: A fresh approach to numerical computing," *SIAM Review*, vol. 59, no. 1, pp. 65–98, 2017. DOI: 10.1137/141000671. eprint: https://doi.org/10.1137/141000671.

[41] M. Ashahrani and N. Almuraysil, *Drdeed.jl: A julia package for demand response and dynamic economic emission dispatch*, https://github.com/mmogib/DRDeed.jl, 2024.

[42] M. Ashahrani and N. Almuraysil, *Tables of data and results*, https://github.com/mmogib/NMSPaperMAY24, 2024.

[43] Y. Y. Haimes, L. S. Lasdon, and D. A. Wismer, "On a bicriterion formulation of the problems of integrated system identification and system optimization," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-1, no. 3, pp. 296–297, 1971, Original formulation of the ε-constraint method for multi-objective optimization. DOI: 10.1109/TSMC.1971.4308298.

[44] H. M. Alshehri *et al.*, "Reducing high energy demand associated with air-conditioning needs in Saudi Arabia," *Energies*, vol. 12, no. 1, p. 87, 2019. DOI: 10.3390/en12010087.

[45] Saudi Electricity Company, "Annual report 2023," Saudi Electricity Company (SEC), Riyadh, Saudi Arabia, Tech. Rep., 2023, Load factor and tariff data for the Eastern Province.

[46] Ministry of Energy, Kingdom of Saudi Arabia, *Peak load and electricity consumption by region (2023)*, Saudi Open Data Portal, https://open.data.gov.sa, Dataset IDs: 004003e8 (Peak Load), 1047dddb (Consumption by Region), 2023.

[47] A. R. Al-Roomi, *Power system optimization: Economic load dispatch*, https://al-roomi.org/power-flow/economic-load-dispatch, Generator cost coefficient data for thermal units, 2015.

[48] U.S. Environmental Protection Agency, "AP-42: Compilation of air pollutant emission factors, volume I: Stationary point and area sources," U.S. EPA, Office of Air Quality Planning and Standards, Tech. Rep., 2024, Chapter 3: Stationary Internal Combustion Sources (gas turbines, oil-fired boilers).